The Fourier Qualia Space:

Interaction, Ambiguity, and Hierarchy in Music Harmony

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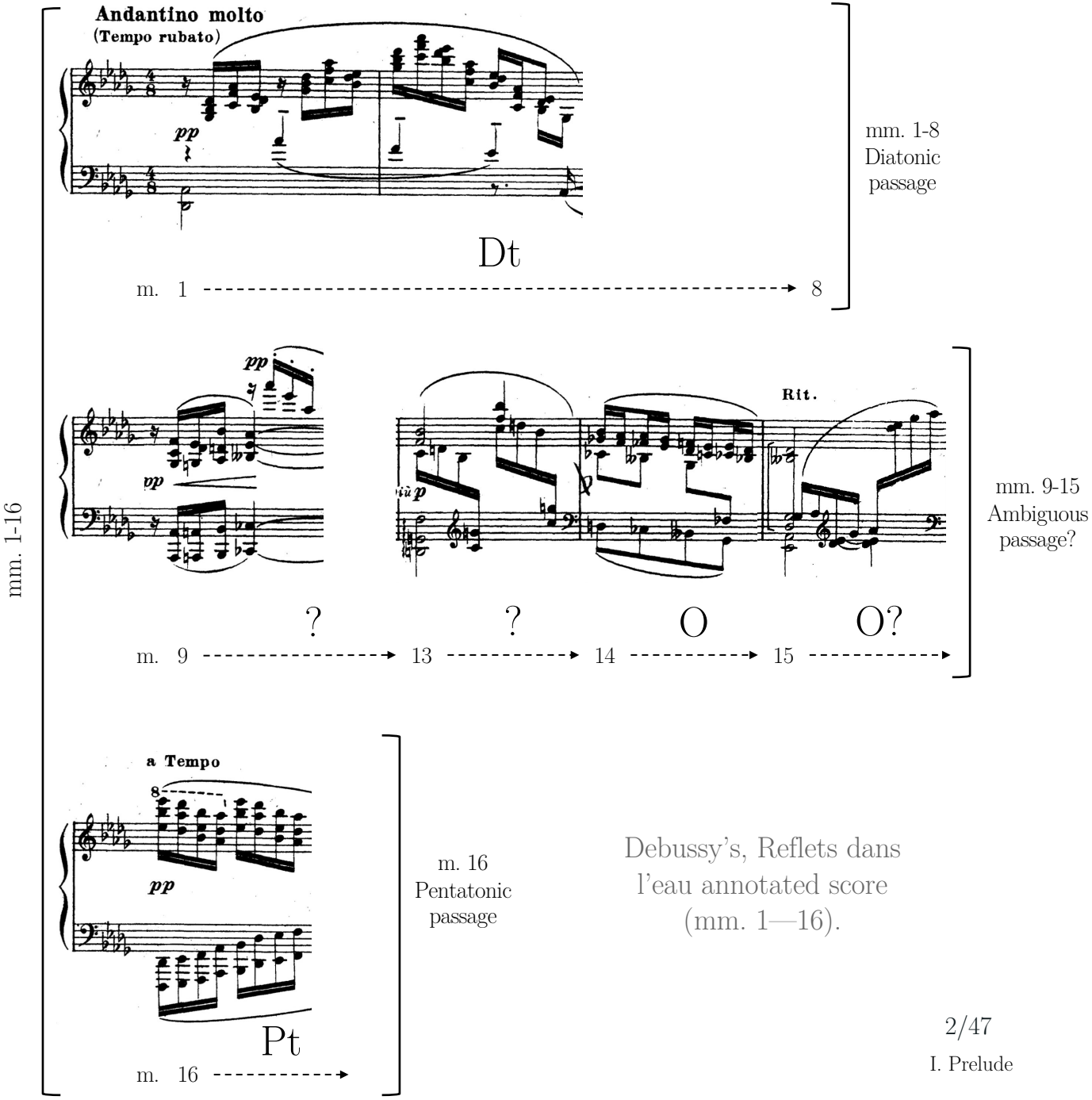
I. Prelude.

Listen.

Debussy, Reflets dans l'eau
mm. 1—16. Listen 1/3.

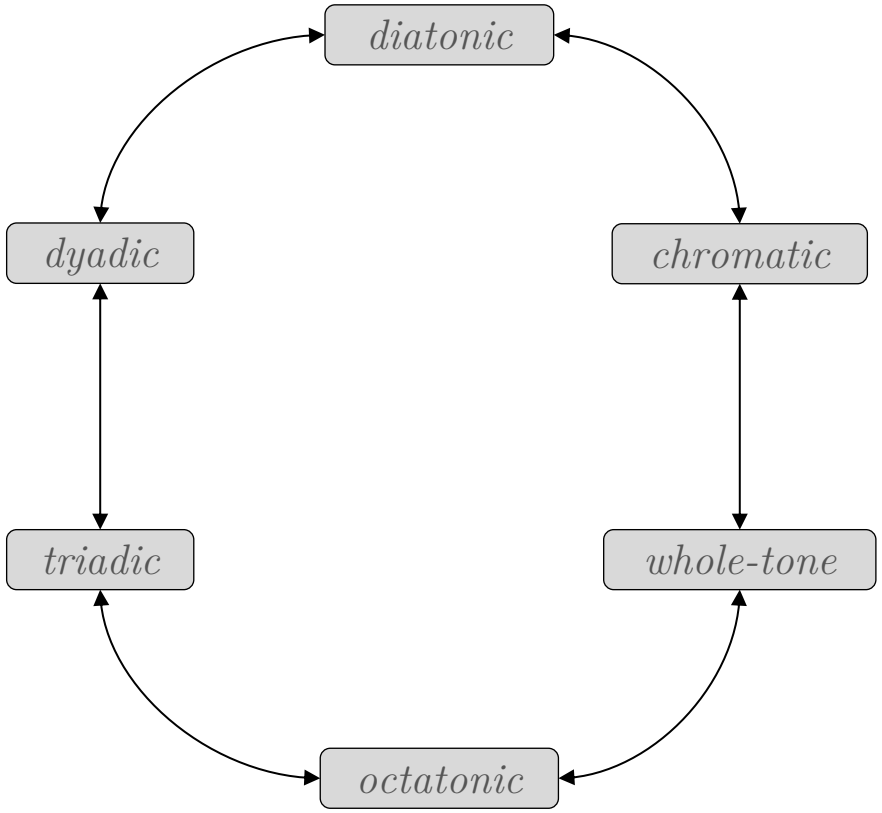
"a series of aquatic chords that piano pedals rendered thoughtful
their dissonant tones formed the shimmering image
of the thin fractures that in life,
my own life or that of others, are there or are missing."

Jorge de Sena, in ***"La Cathédrale Engloutie," by Debussy***



Debussy's, Reflets dans l'eau annotated score (mm. 1—16).

1. Measuring the vocabulary.



Immediately perceived qualities, routinely composed with, yet resistant to systematic measurement.

Listeners hear and harmonic qualities and composers deploy them deliberately across repertoire. Yet no shared quantitative structure locates them.

2. Three lineages, three remainders.

1 **Functional Harmony**
Rameau → Riemann → Schenker

A gravitational system tuned to tonic–dominant polarity. Illuminates Bach, Mozart, Beethoven, or Schubert. Collapses on Debussy's parallel chords and Schoenberg's emancipated dissonance.

2 **Set Theory**
Forte and successors

Built precisely to handle post-tonal pitch collections. Classifies sets exhaustively but discards circular order and rotational alignment.

3 **DFT Lineage**
Quinn, Amiot, Yust

A quantitative breakthrough but stops short of a full quality space.

3. Three gaps.



Interaction & Motion

Vocabulary describes static states. It is silent about how a piece moves between qualities.



Ambiguity

Passages in which several qualities coexist without any one dominating are not treated as a compositional resource. No metric exists for qualia ambiguity.



Multidimensionality

Existing spaces handle two qualities at once at most. Music routinely traverses several qualities in parallel and even superimposes them.

4. Three questions from one enquiry.

> **Central enquiry:** Do DFT-derived geometric properties of pitch-class sets supply a measurable framework that captures harmonic-quality features thus revealing systematic principles that bridge the tonal–atonal divide?



(i) Do geometric representations of motion through the quality-space expose structural connections hidden to traditional analysis?

→ *addresses the Interaction & Motion and the Ambiguity gaps*



(ii) What evolutionary patterns emerge when this framework is applied computationally to a large spanning corpus?

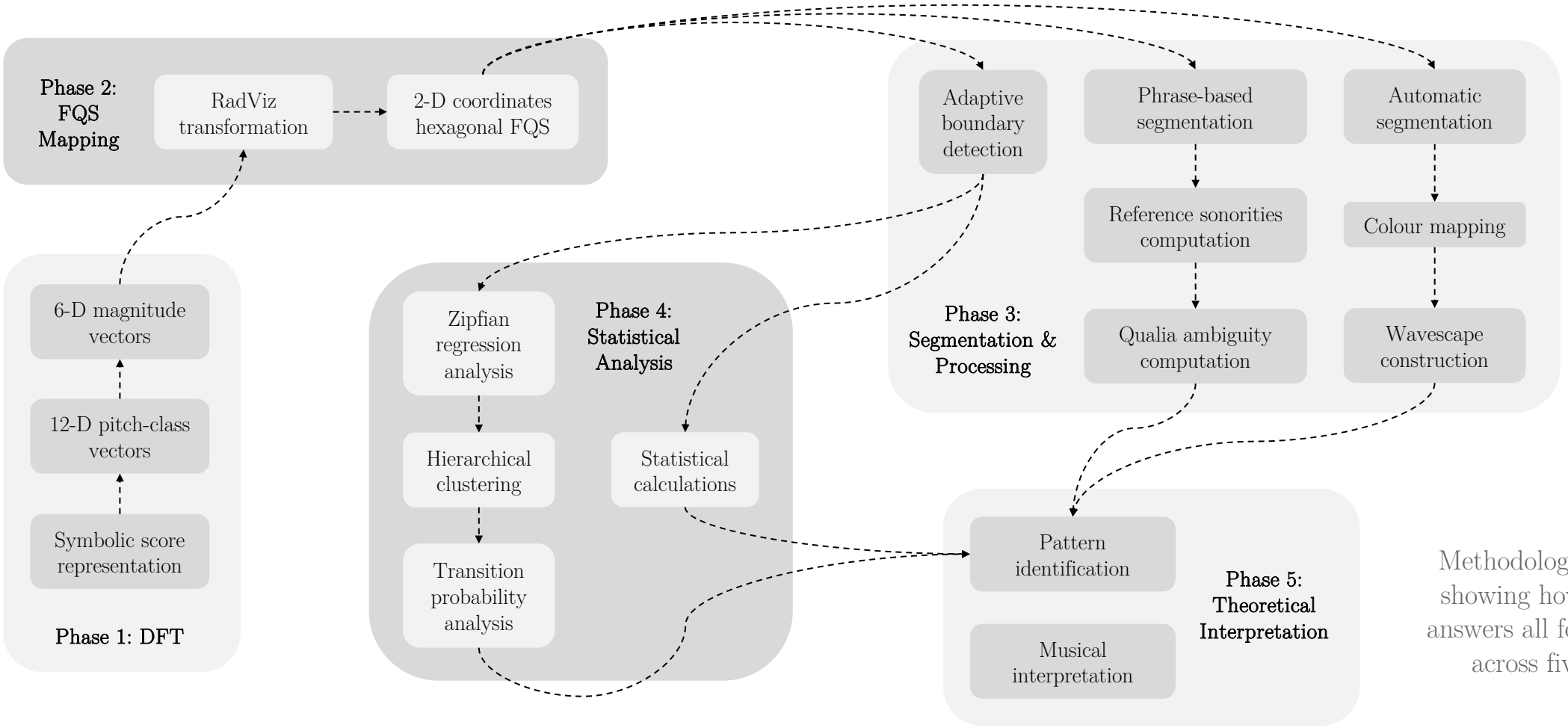
→ *enabled by restoring Multidimensionality*



(iii) Do post-tonal repertoires display statistical patterns with grammar-like properties?

→ *enabled by restoring all three gaps*

5. Five phases in one pipeline.



Methodological pipeline showing how the thesis answers all four questions across five stages.

6. Agenda.

I. Prelude — A listening prelude

II. Foundations — Phenomenological, mathematical, and spatial music theory

III. Advancement — The Fourier Qualia Space

IV. Application 1 — Motion, contrapuntal linearity, and nondirected ambiguity

V. Application 2 — Qualia hierarchies through wavescapes

VI. Application 3 — A linguistic turn

VII. Postlude — Synthesis and conclusion

II. Foundations.

Phenomenological, mathematical, and spatial music theory.

"if it all is organized as a varied image
of an idea bereft of meaning;
if it all sounds like the freedom of logical accidents
for groups, and great numbers, and proportions
to recognize as necessary;"

Jorge de Sena, in ***Bach: Goldberg Variations***

1. Qualia reconsidered.

This Thesis

Reconceptualises "Qualia" to encompass all four dimensions.

Harmonic-quality reconceived across four dimensions — treated as *harmonic qualia* to encompass all four:

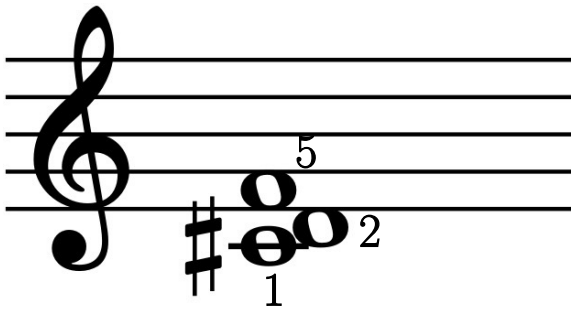
- spatial-geometric — quality as position in space (Quinn, 2006–7);
- temporal-accumulative — macroharmony (Tymoczko, 2011);
- hierarchical — quality across structural levels (Viaccoz et al., 2023);
- phenomenological — the measurements correspond to something we perceive.

Note on perception: not a perceptual study. What the Fourier transform measures carries perceptual implications — these are not tested experimentally here.

2.1. Pitch-class sets as binary vectors.

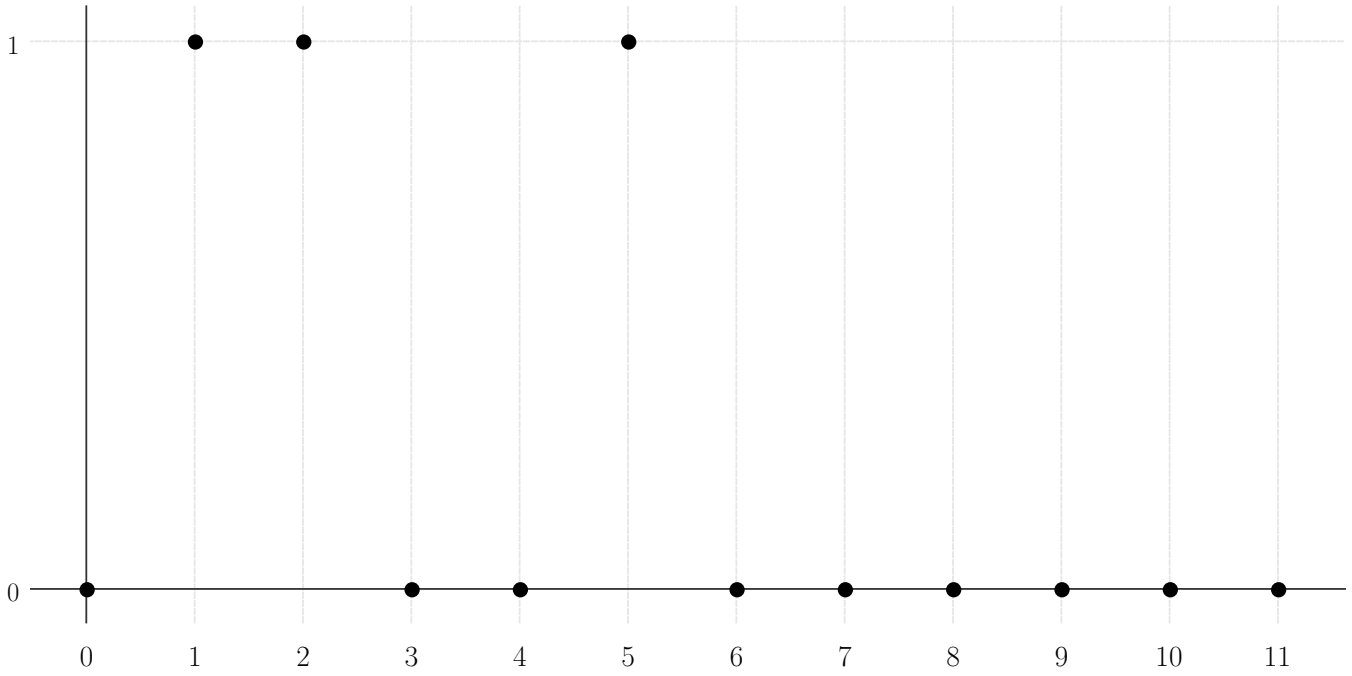
- Every pitch-class set becomes a 12-element binary vector over the pitch-classes 0–11.
- Example:

  {C♯, D, F} →
  {1, 2, 5} →
  (0,1,1,0,0,1,0,0,0,0,0,0).



Pitch-class set {1, 2, 5}.

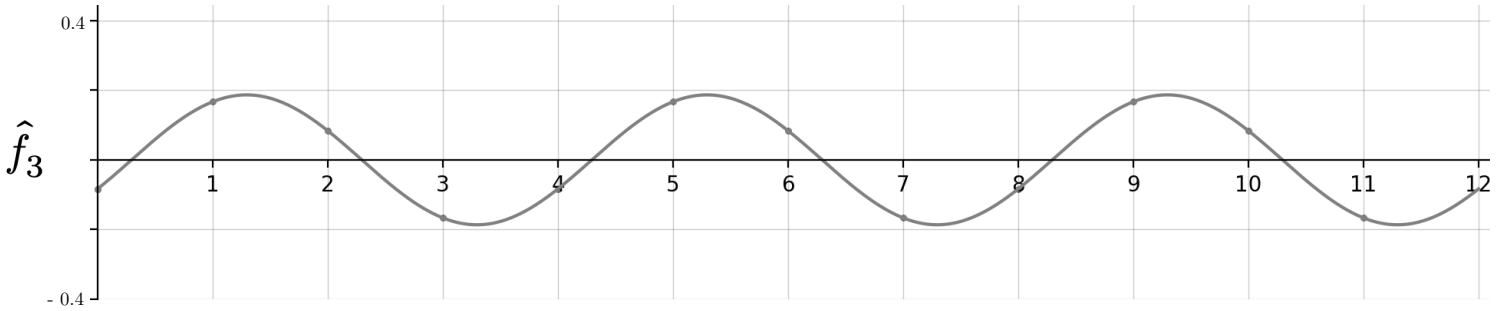
- Minimally abstract—a concrete object.
- This representation is the input for every step that follows.



Binary encoding of pitch-class set {1, 2, 5} as a 12-element vector.

2.2. DFT decomposition of the pitch-class vector.

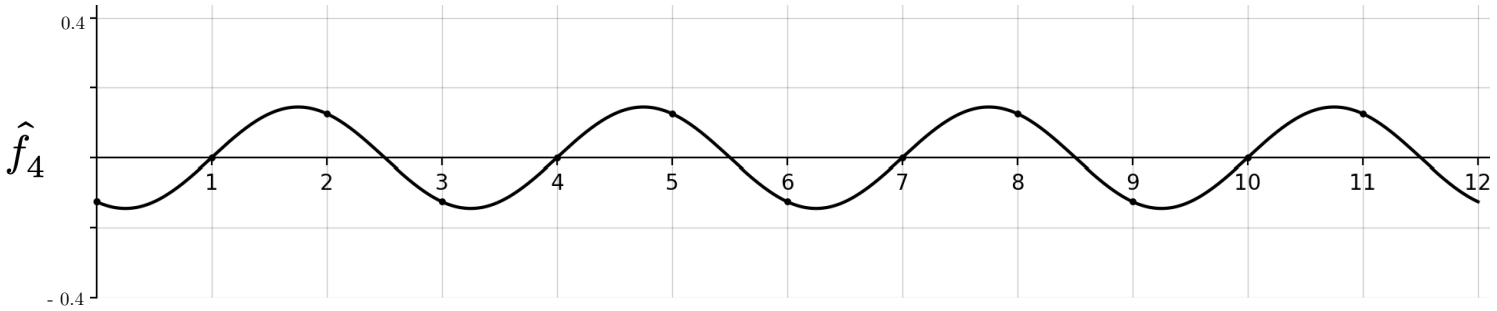
The DFT decomposes any pitch-class vector into **twelve coefficients** (represented here as sinusoids) $\hat{f}_{0—11}$. Each coefficient **captures a specific periodicity** in the pitch-class distribution. The original pitch-class vector becomes mathematically reconstructible from its decomposition.



Sinusoid of the DFT coefficient 3 calculated for pitch-class set {1, 2, 5}.

2.2. DFT decomposition of the pitch-class vector.

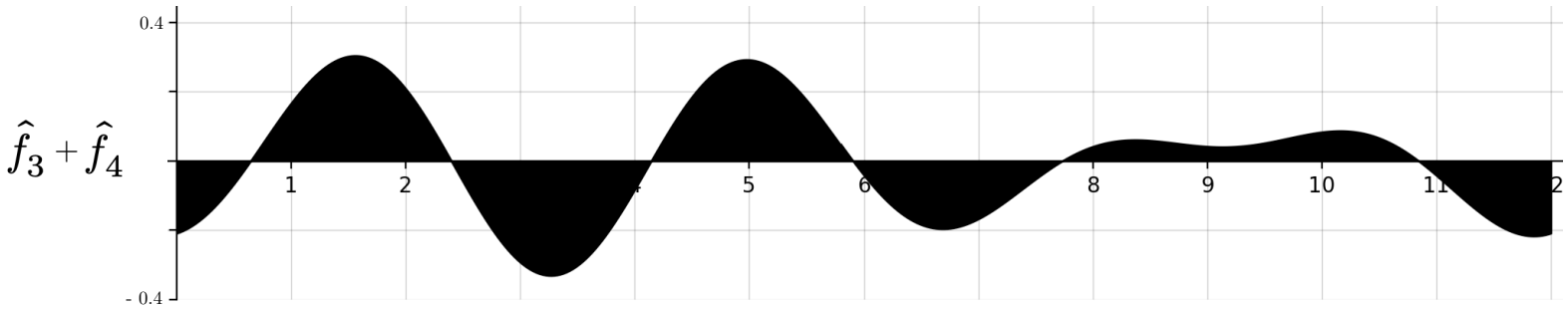
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Sinusoid of the DFT coefficient 4 calculated for pitch-class set {1, 2, 5}.

2.2. DFT decomposition of the pitch-class vector.

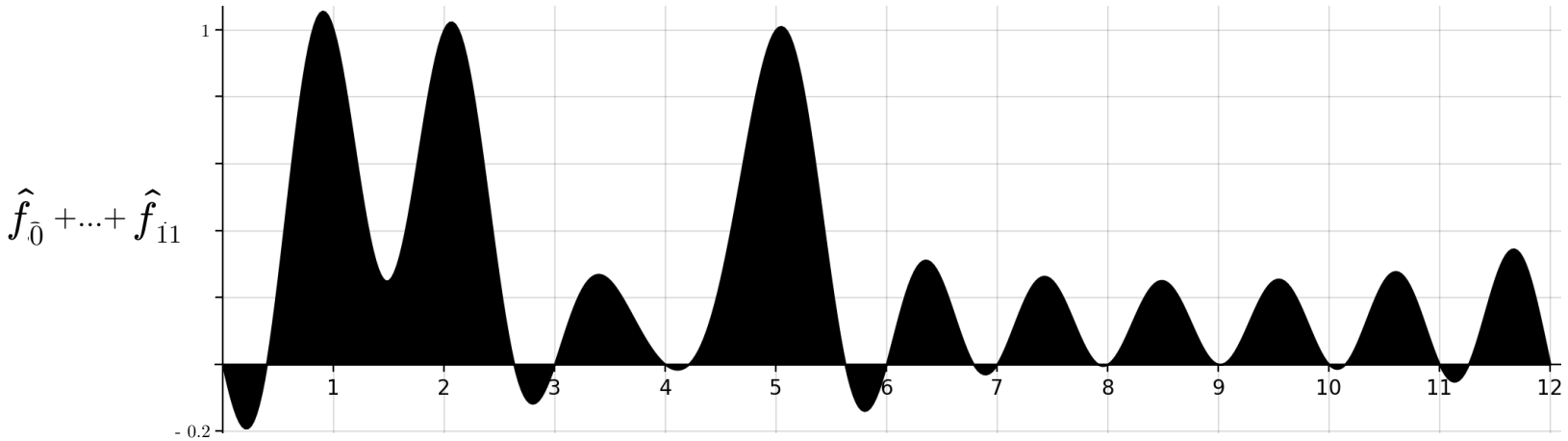
The DFT decomposes any pitch-class vector into **twelve coefficients** (represented here as sinusoids) $\hat{f}_{0—11}$. Each coefficient **captures a specific periodicity** in the pitch-class distribution. The original pitch-class vector becomes mathematically reconstructible from its decomposition.



Sinusoid of the sum of DFT coefficients 3 and 4 calculated for pitch-class set {1, 2, 5}.

2.2. DFT decomposition of the pitch-class vector.

The DFT decomposes any pitch-class vector into **twelve coefficients** (represented here as sinusoids) $\hat{f}_{0—11}$. Each coefficient **captures a specific periodicity** in the pitch-class distribution. The original pitch-class vector becomes mathematically reconstructible from its decomposition.

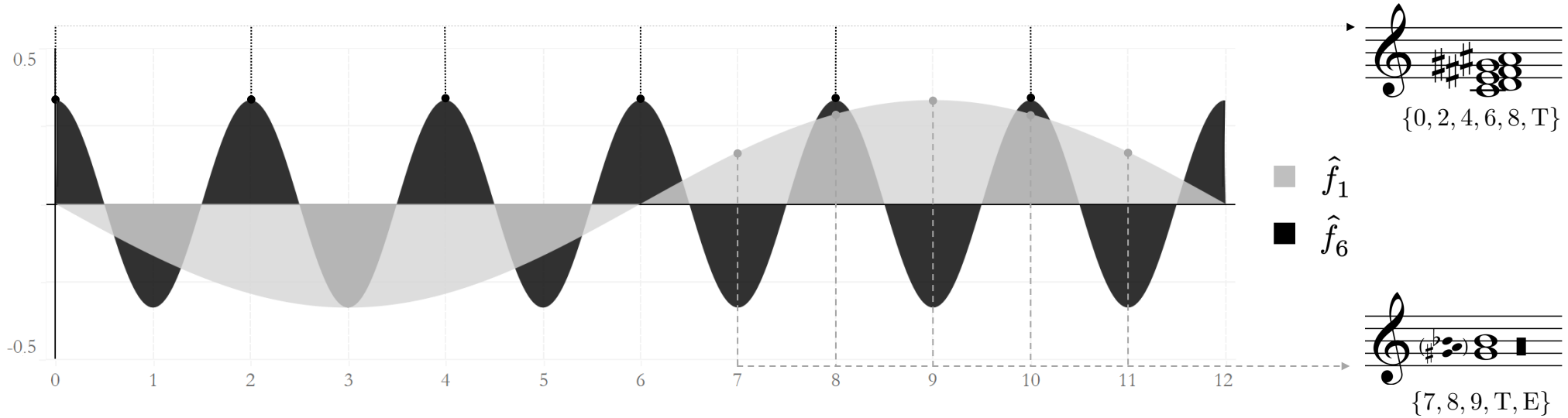


Sinusoid of the DFT coefficient 4 calculated for pitch-class set {1, 2, 5}.

2.3. Six non-trivial coefficients, six harmonic qualities.

Six non-trivial coefficients map to six qualities:

- $\hat{f}_1 \rightarrow$ chromaticity (one period)
- $\hat{f}_6 \rightarrow$ whole-tone quality (six periods)

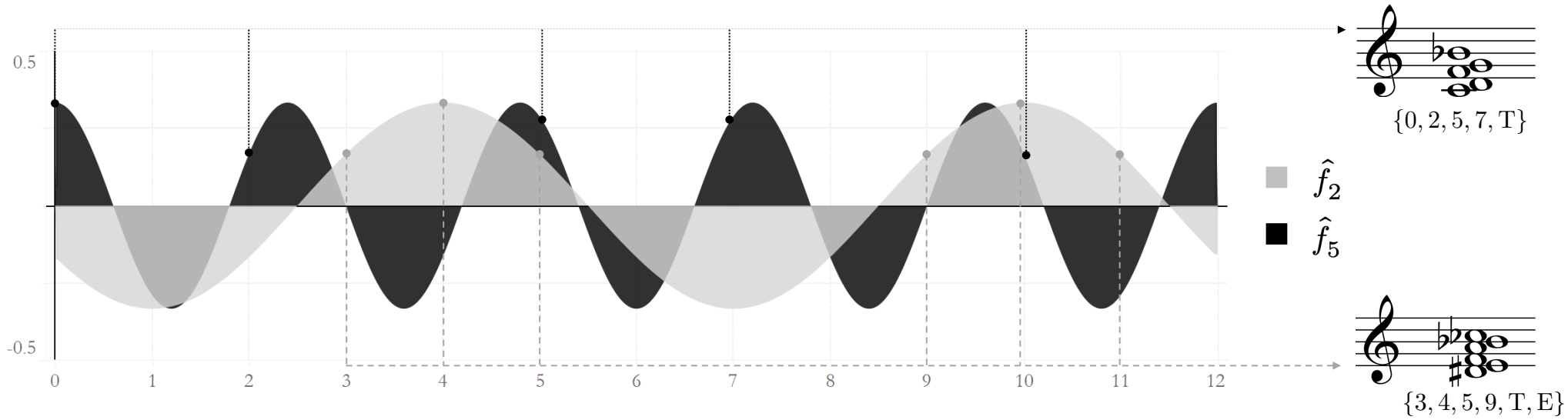


Two DFT coefficient sinusoids paired by period: chromaticity and whole-tone quality.

2.3. Six non-trivial coefficients, six harmonic qualities.

Six non-trivial coefficients map to six qualities:

- $\hat{f}_1$ → chromaticity (one period)
- $\hat{f}_2$ → dyadicity (two periods)
- $\hat{f}_6$ → whole-tone quality (six periods)
- $\hat{f}_5$ → diatonicity (five periods)

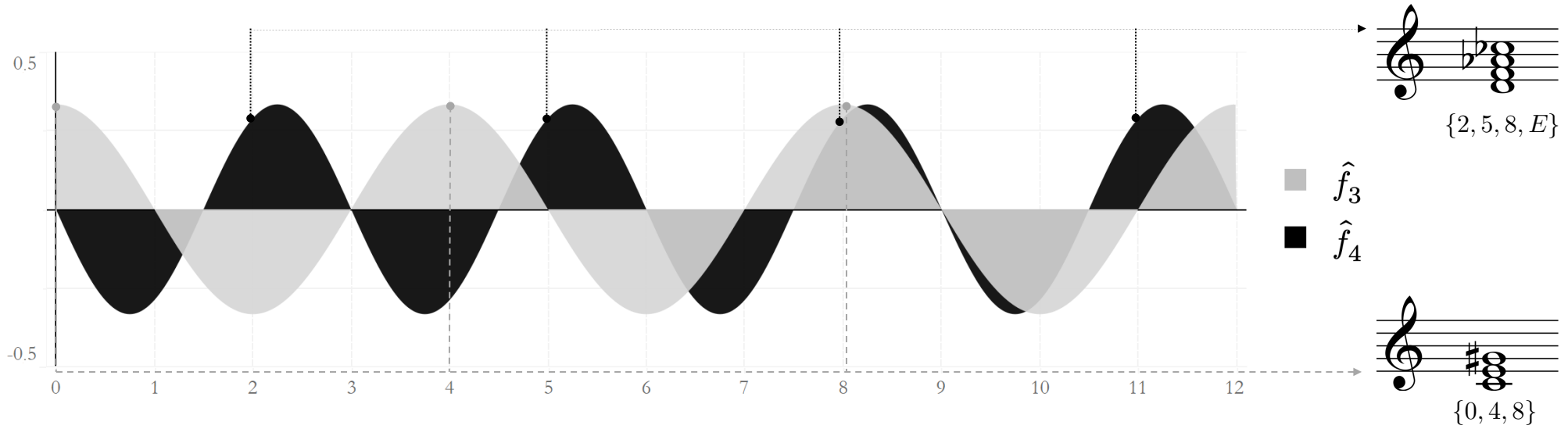


Two DFT coefficient sinusoids paired by period: dyadicity and diatonicity.

2.3. Six non-trivial coefficients, six harmonic qualities.

Six non-trivial coefficients map to six qualities:

- $\hat{f}_1$ → chromaticity (one period)
- $\hat{f}_2$ → dyadicity (two periods)
- $\hat{f}_3$ → triadicity (three periods)
- $\hat{f}_6$ → whole-tone quality (six periods)
- $\hat{f}_5$ → diatonicity (five periods)
- $\hat{f}_4$ → octatonicity (four periods)

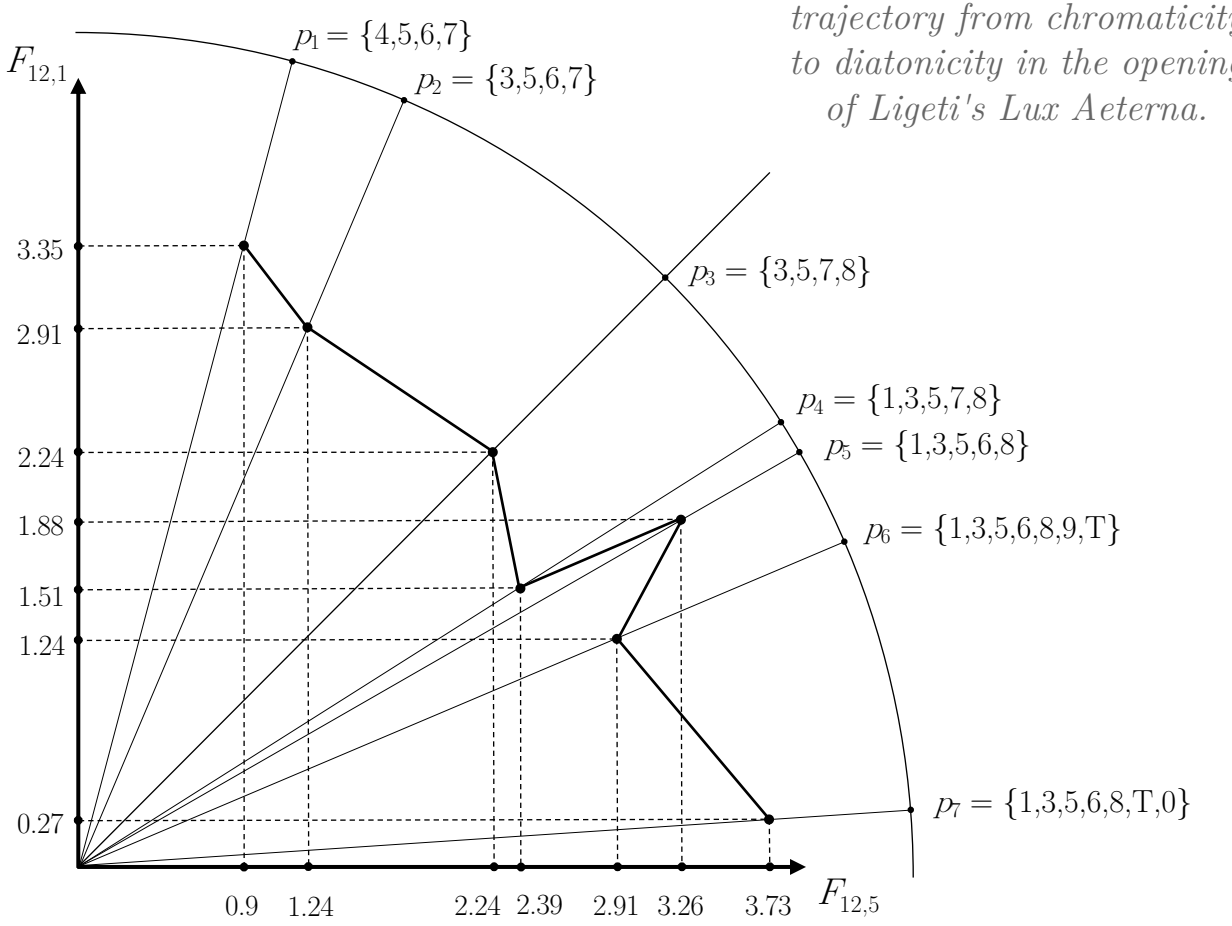


Two DFT coefficient sinusoids paired by period: triadicity and octatonicity.

3. From DFT magnitudes to a quality space.

Quinn (2006–07) opened the door: DFT magnitudes^ as a quality space, but his trajectory lives in a plane defined by two salient coefficients only.

The FQS extends Quinn's quality space from 2 to 6 dimensions.



Quinn's Quality Space trajectory from chromaticity to diatonicity in the opening of Ligeti's Lux Aeterna.

III. Advancement.

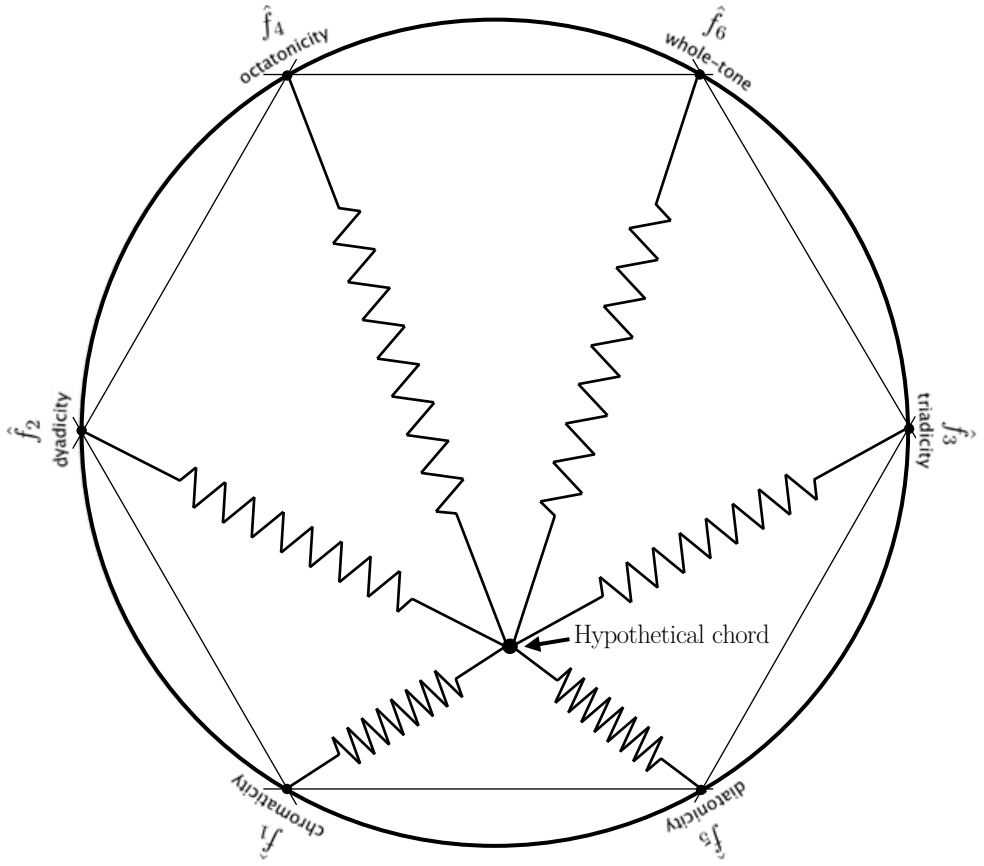
The Fourier Qualia Space.

"the virtual thought became
an evidence: became concrete.
A concreteness—and this is the awe—made of exterior things
equal to the abstractness inner things could have."

Jorge de Sena, in ***Bach: Goldberg Variations***

1. Hexagonal anchoring and mechanical analogy.

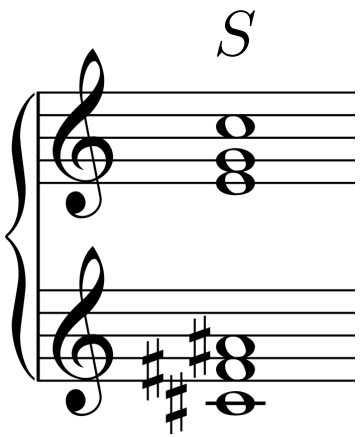
In the FQS, the six DFT coefficient labels sit at the vertices of a regular hexagon. Picture a chord pulled by **six springs** toward the six hexagon anchors. Spring stiffness = DFT magnitude of the corresponding coefficient.



RadViz anchor positioning in the FQS and its spring model of forces.

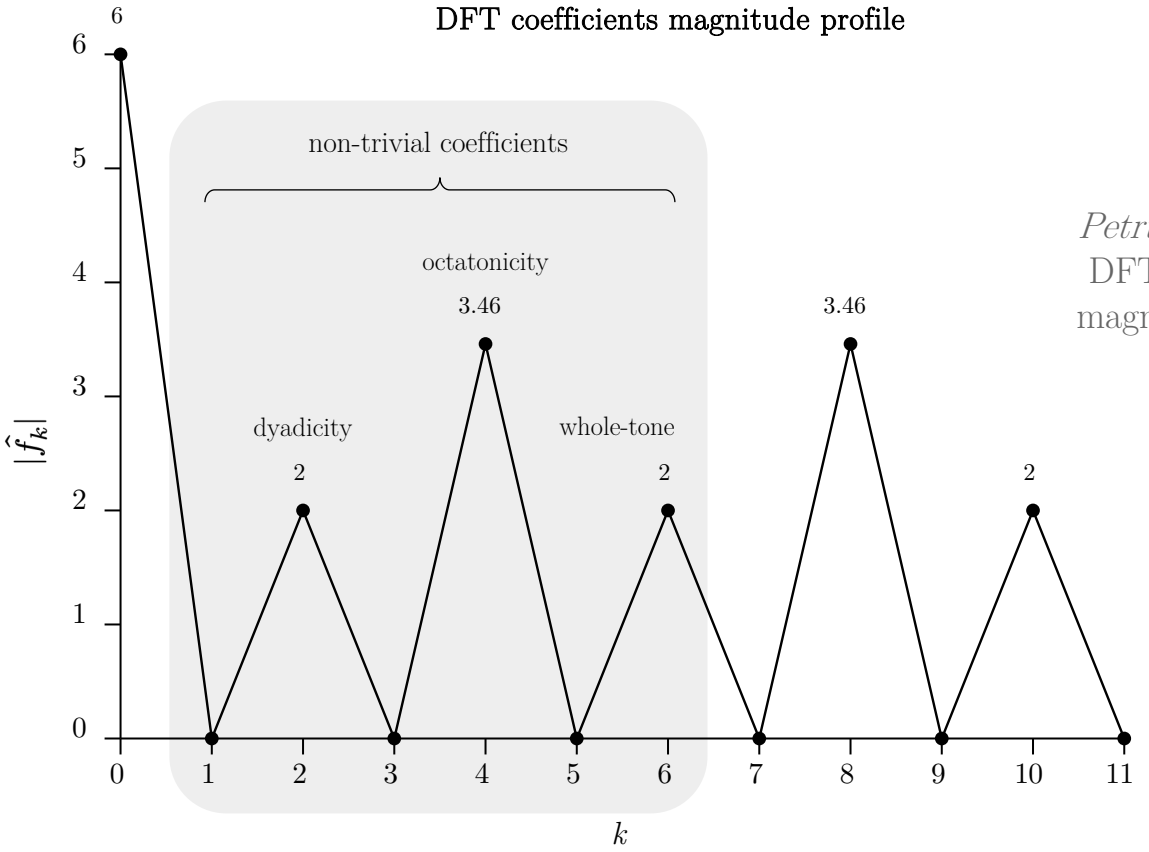
2. *Petrushka* chord: a worked example.

- Petrushka chord: set class 6-30[013679].
- Comprised of juxtaposed major chords a tritone apart.



Petrushka chord.

- Pulled only by octatonicity, dyadicity, and whole-tone.



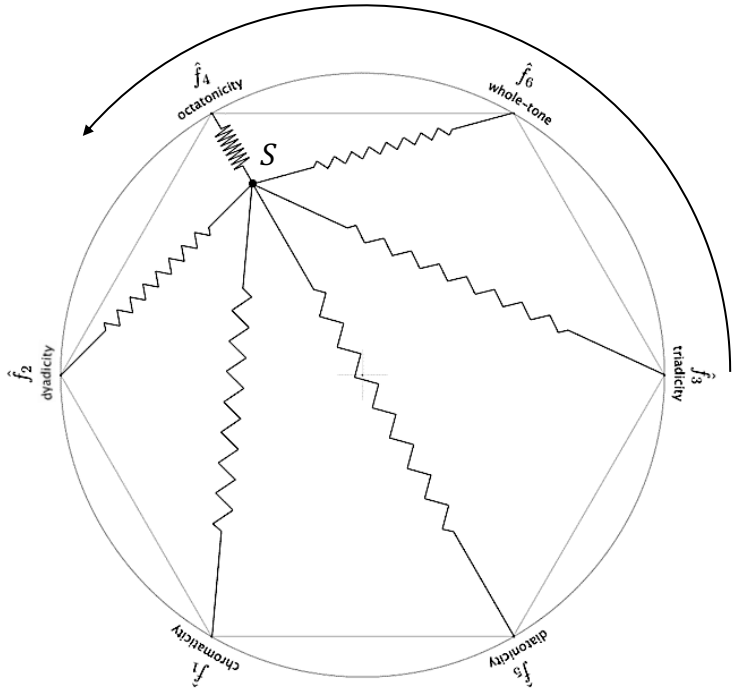
Petrushka's chord DFT coefficients magnitude profile.

2. *Petrushka* chord: a worked example.

- However, different arrangements of DFT coefficients as anchors produce distinct positionings for the same set class.

DFT coefficients order:

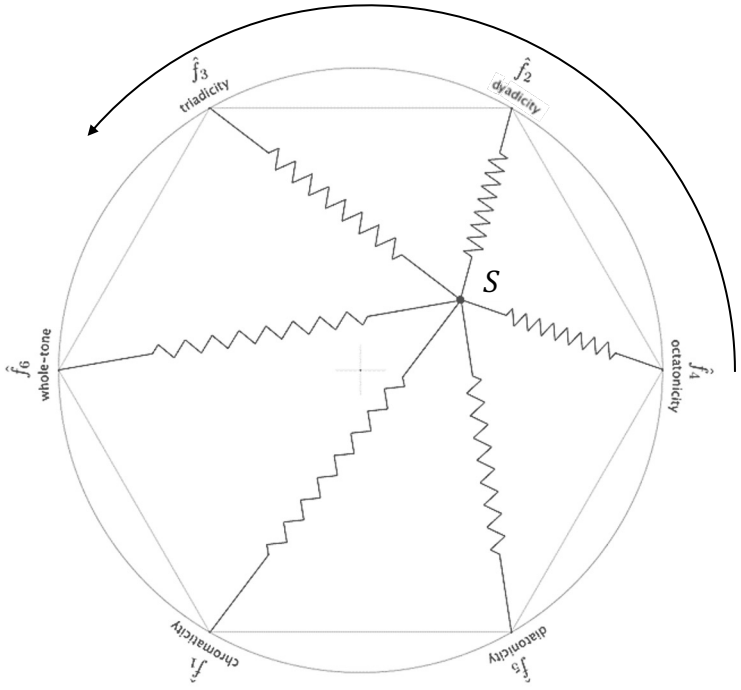
3, 6, 4, 2, 1, and 5

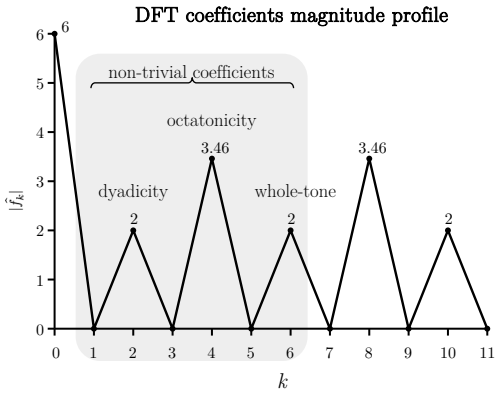


≠

DFT coefficients order:

4, 2, 3, 6, 1, and 5





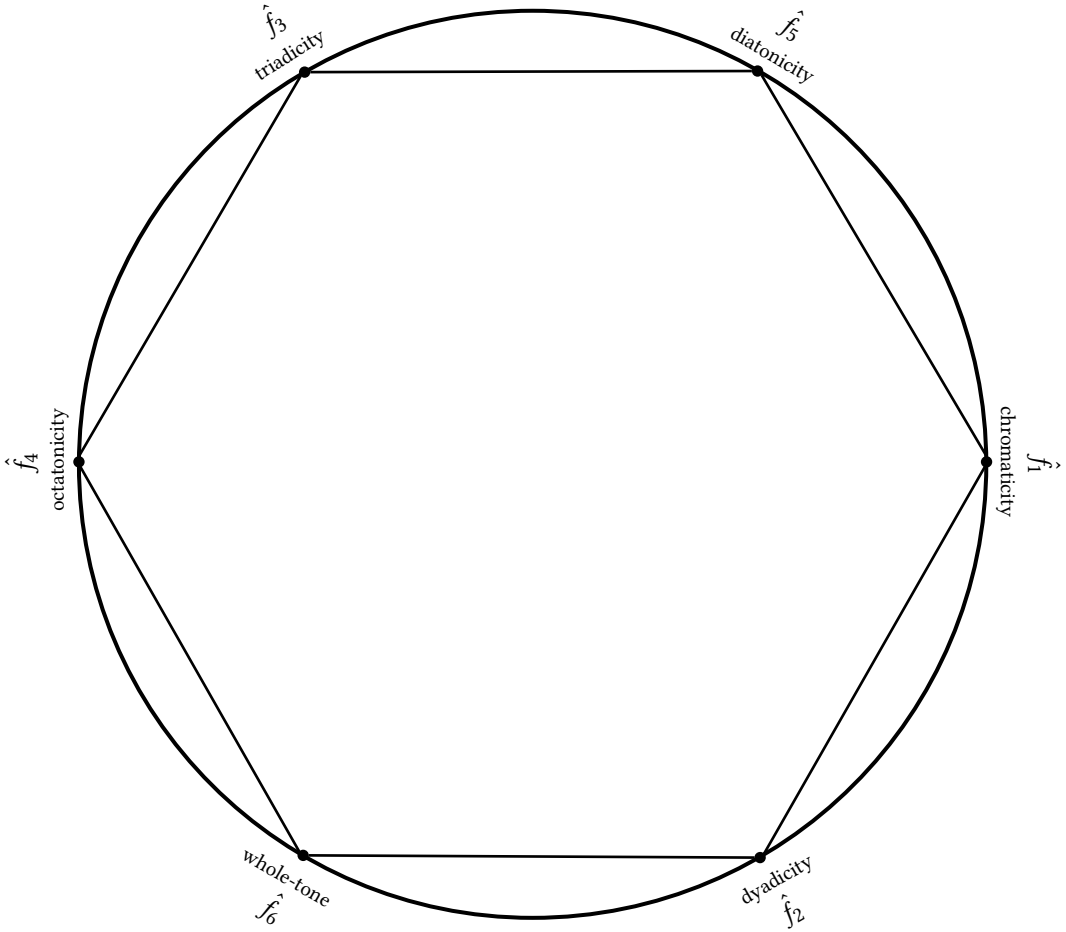
Stravinsky's *Petrushka* chord projected into the FQS with two distinct arrangements of DFT coefficients.

3. Finding the optimal ordering.

The optimal ordering is $\hat{f}_3, \hat{f}_5, \hat{f}_1, \hat{f}_2, \hat{f}_6, \hat{f}_4$. This arrangement is not decorative! It is the result of an optimisation process that maximises:

1. Set class dispersion;

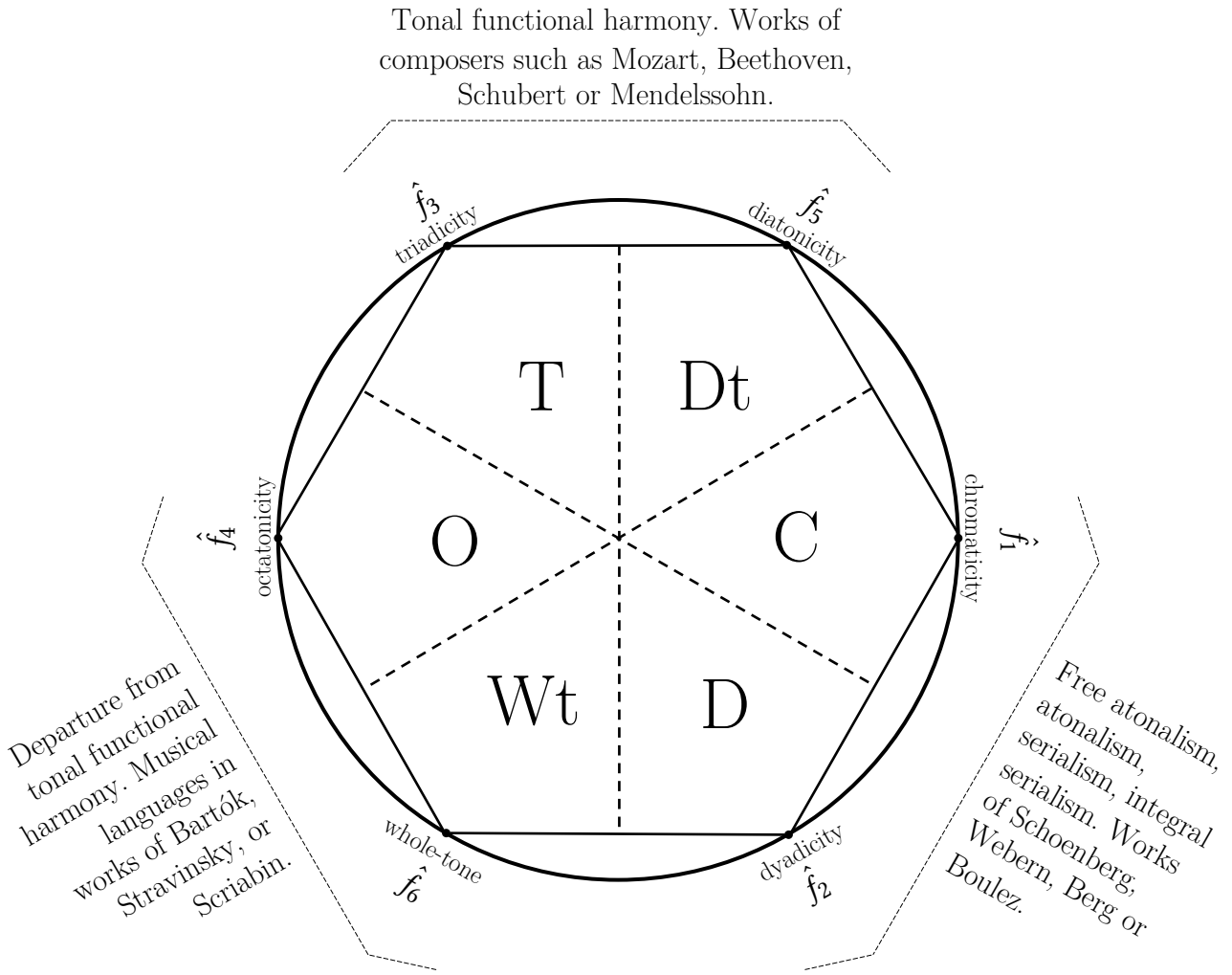
2. Pearson correlation between the FQS and 6-D space.



Optimal DFT coefficients ordering as FQS's anchors.

4. Regions, meta-regions, and the geography of style.

The FQS naturally partitions into **six primary regions**, one per quality, and spontaneously divides into three pairs of adjacent qualities that correspond to broad historical epochs of harmonic practice.



Primary regions and meta-regions over the FQS.

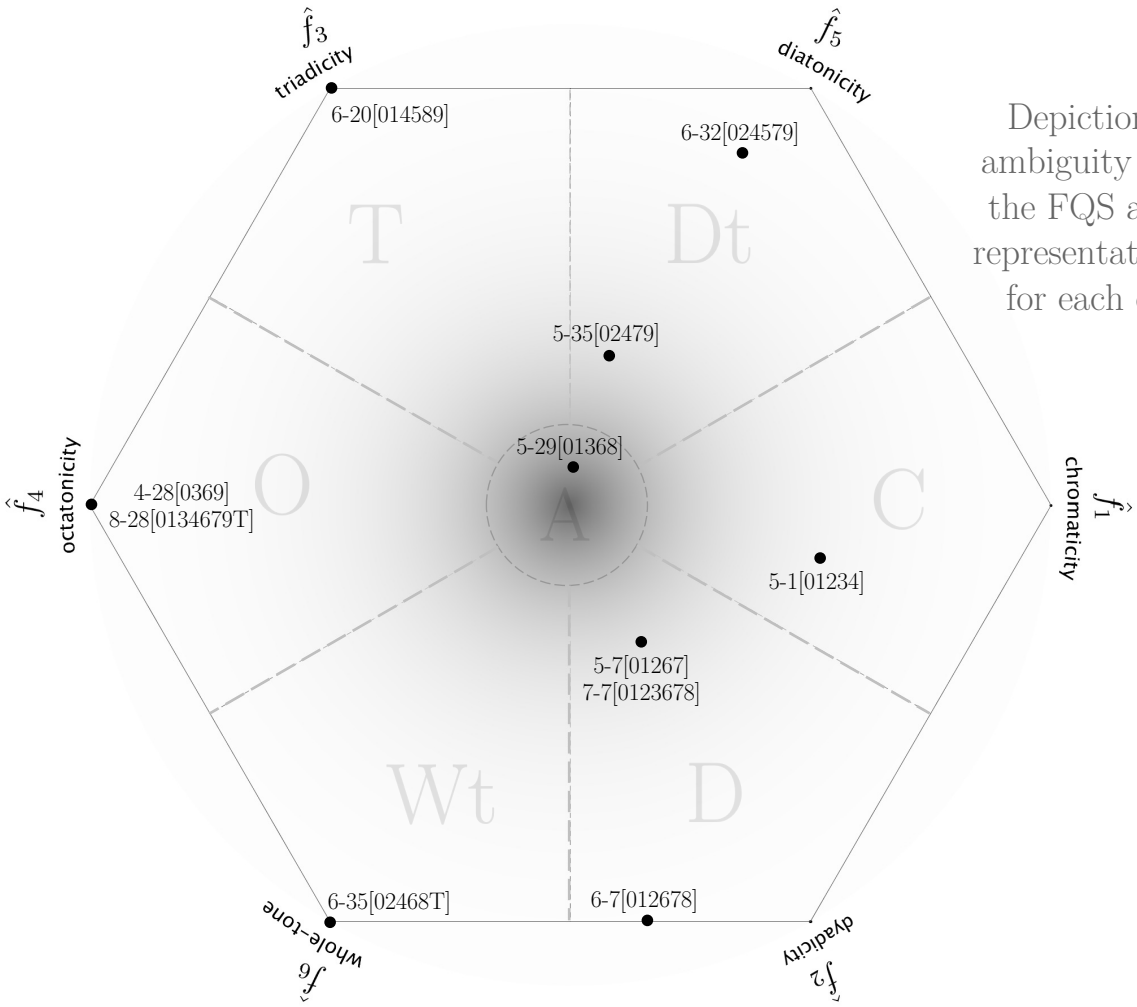
5. Qualia ambiguity (η) as a seventh quale.

The FQS's centre stands as a singular point where no quality predominates:

- A seventh region thus denotes qualia ambiguity.
- η quantifies positional indefiniteness in the FQS. Five benchmarks calibrate it from zero to near one.

Set group	Composers	η
α	Adjacent anchors.	0
β	Opposite anchors	0.347
γ	Mixed sets	0.453
δ	Diatonic phrase	0.713
ϵ	Random sets	0.772

Calibration strip for η from pure qualities to random sets.



Depiction of the ambiguity region in the FQS and some representative pcsets for each quality.

IV. Application 1.

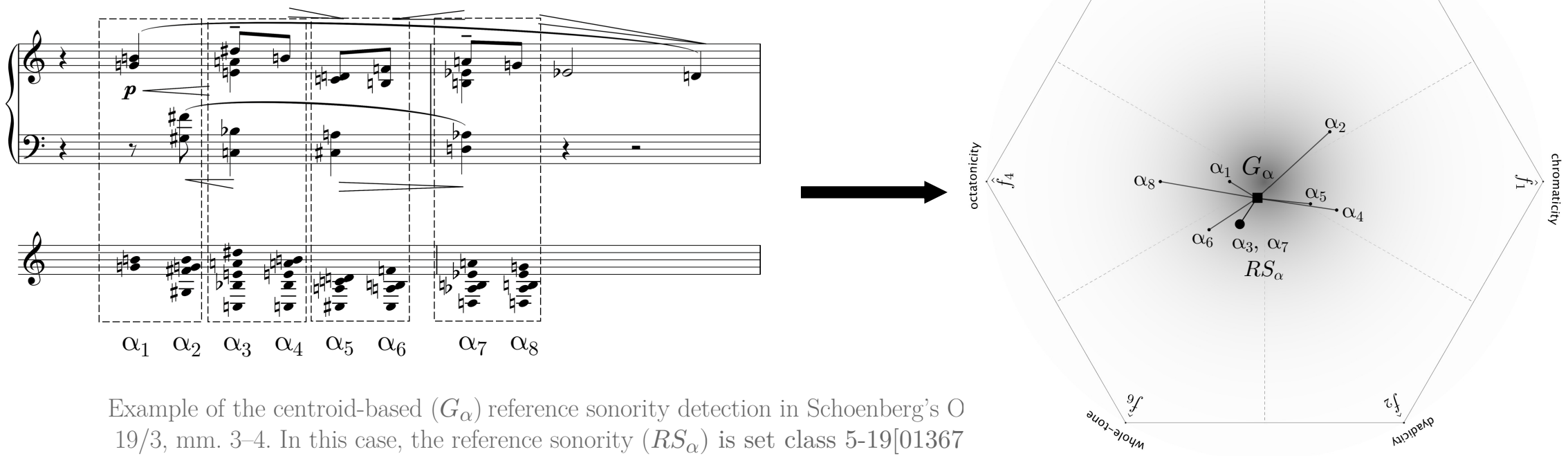
Motion, contrapuntal linearity, and nondirected ambiguity.

"How can such taut strings
throb so passionately that
with the pain of gestures going in arcs
and returning in fluid counterpoint
the weave of seamless music
is like a sea of somber light
into which this furious rigor plunges'"?

Jorge de Sena, in **"Verklärte Nacht," by Schoenberg**

1.From Mailman's RHA to centroid-based reference sonorities.

Thesis adaptation of Mailman's (2015) Representational Hierarchy Association model: (1) **segmentation criterion** = musical phrase; (2) **representative selection criterion** = centroid-based geometric selection in the FQS; and (3) **association arena** = the FQS itself. **The centroid is democratic; the reference sonority is an actual chord, not an abstraction.**



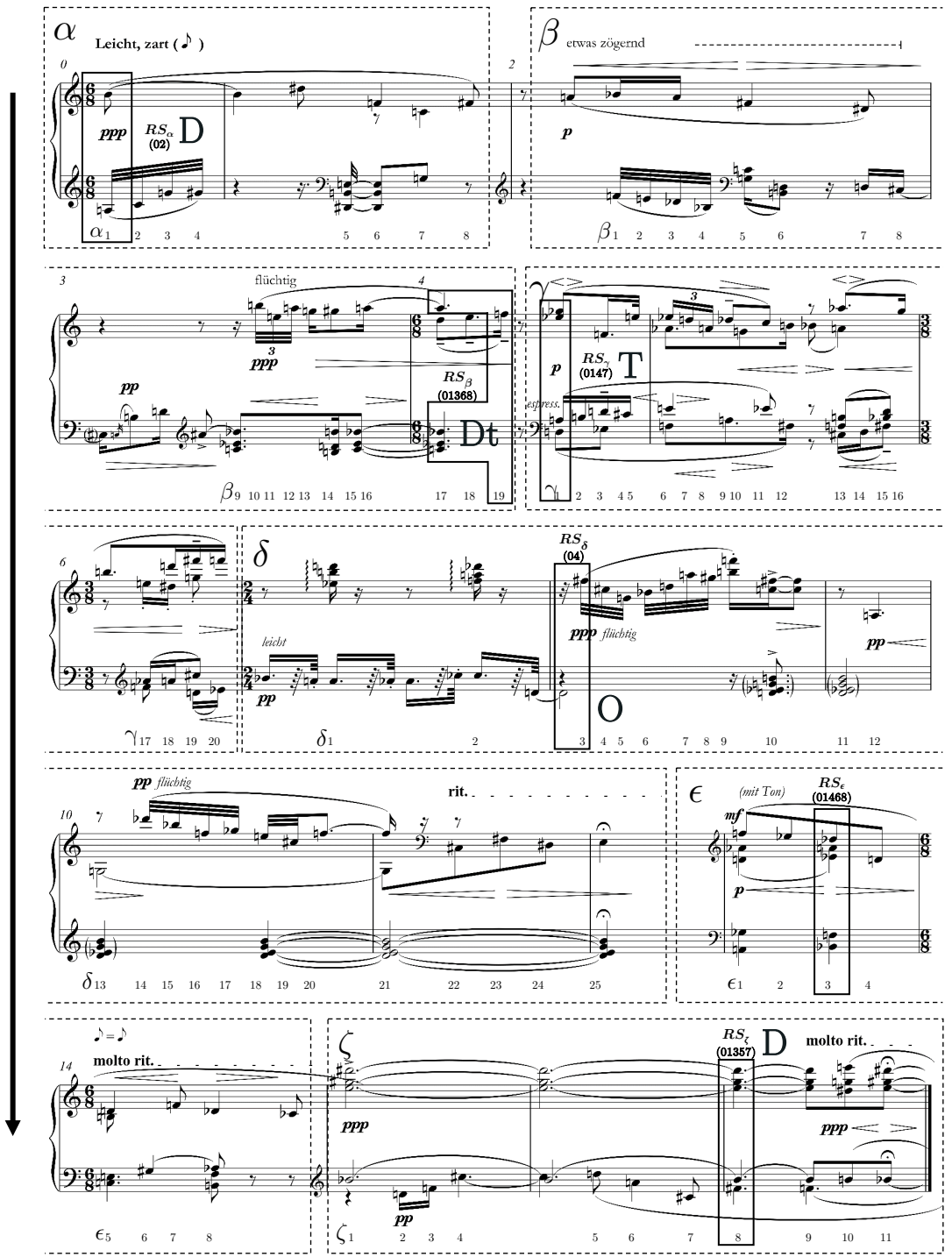
Example of the centroid-based (G_α) reference sonority detection in Schoenberg's O 19/3, mm. 3–4. In this case, the reference sonority (RS_α) is set class 5-19[01367

Listen.

Schoenberg, Op. 19, No. 1, Six Little Piano Pieces

Linear parameters:

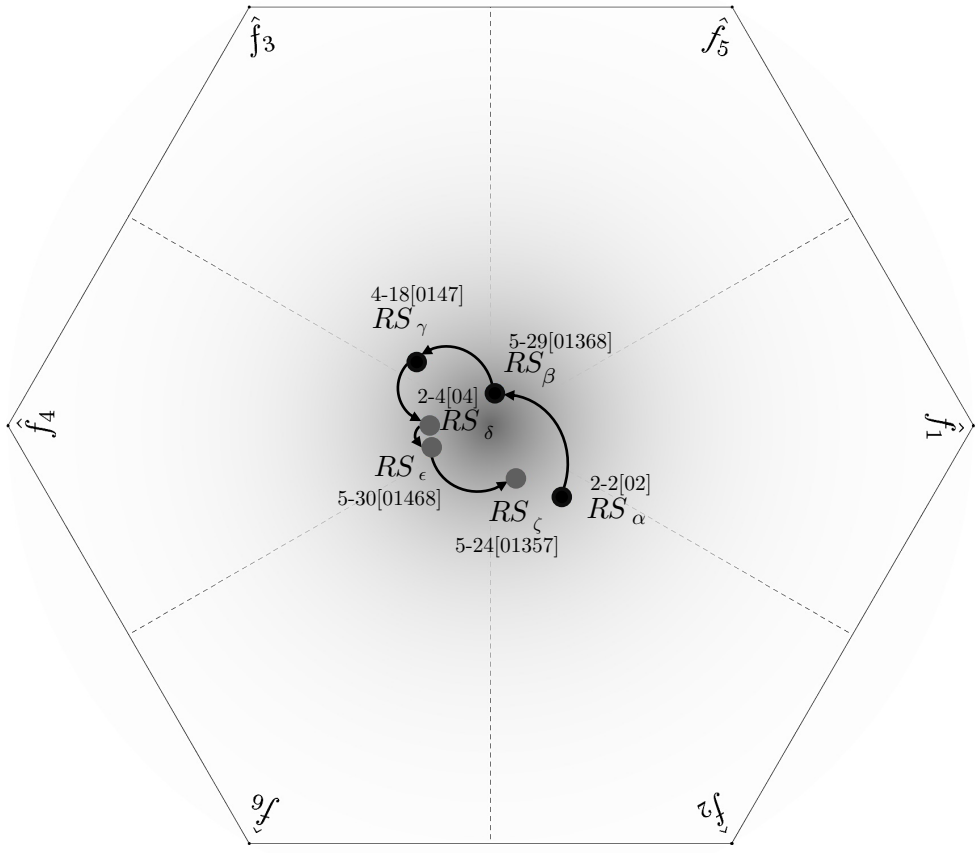
- Less ambiguous meter.
- Texture simplification.
- Clearer stepwise motion.



Schoenberg's, Six Little Piano Pieces, Op. 19/1, annotated score.

2. Schoenberg's Op. 19/1: a qualia trajectory.

- Six phrases → six reference sonorities (α–ζ).
- Anticlockwise circular motion: dyadicity → diatonicity → triadicity → octatonicity → octatonicity → dyadicity.

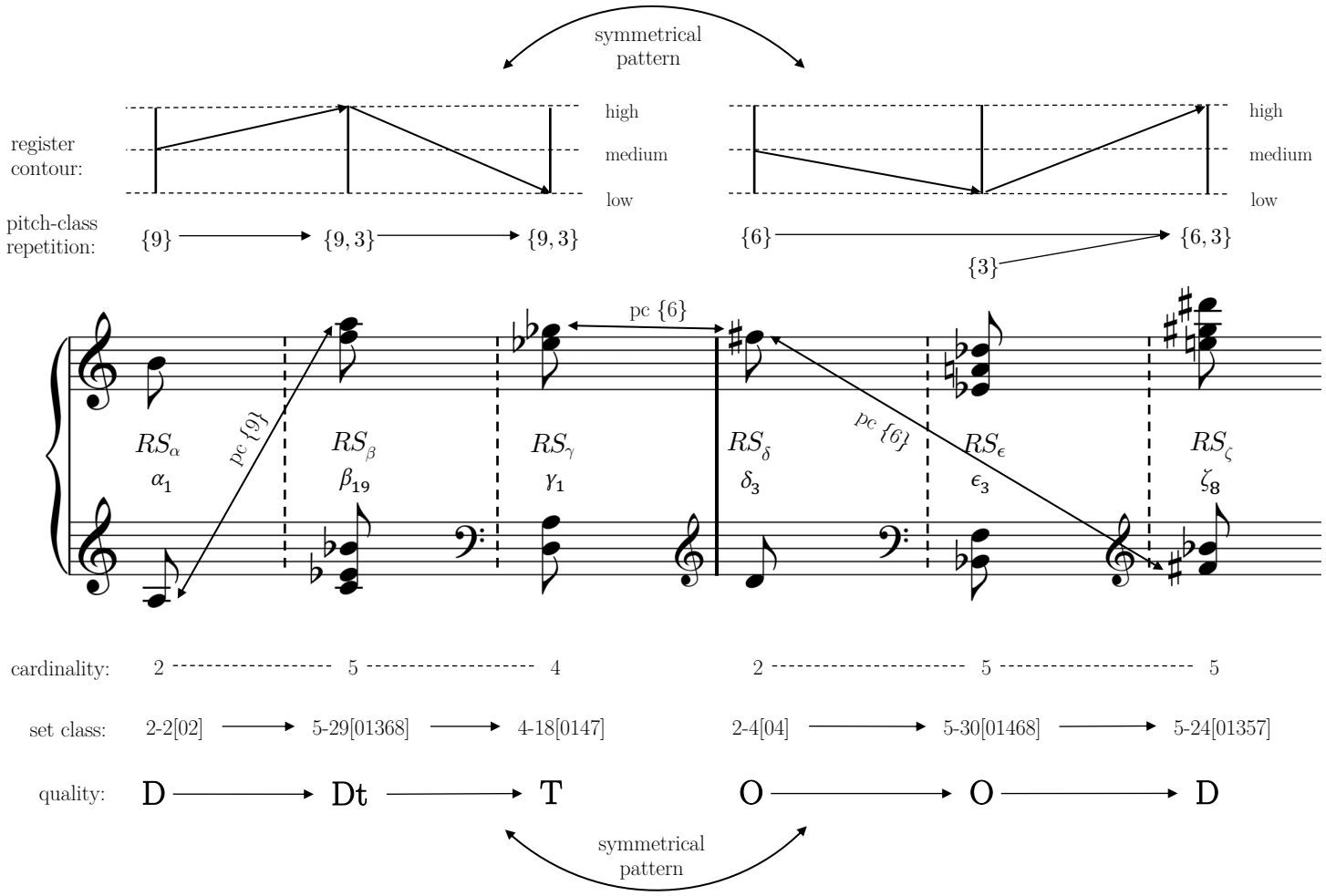


Progression of reference sonorities in Op. 19/1, plotted in the FQS.

2. Schoenberg's Op. 19/1: a qualia trajectory.

- Six phrases → six reference sonorities (α–ζ).
- Anticlockwise circular motion: dyadicity → diatonicity → triadicity → octatonicity → octatonicity → dyadicity.
- Bipartite symmetry: first half mid–high–low, second half mid–low–high; dyadicity brackets the piece.

The FQS produces a formal reading grounded in geometry, not in set-class catalogues.



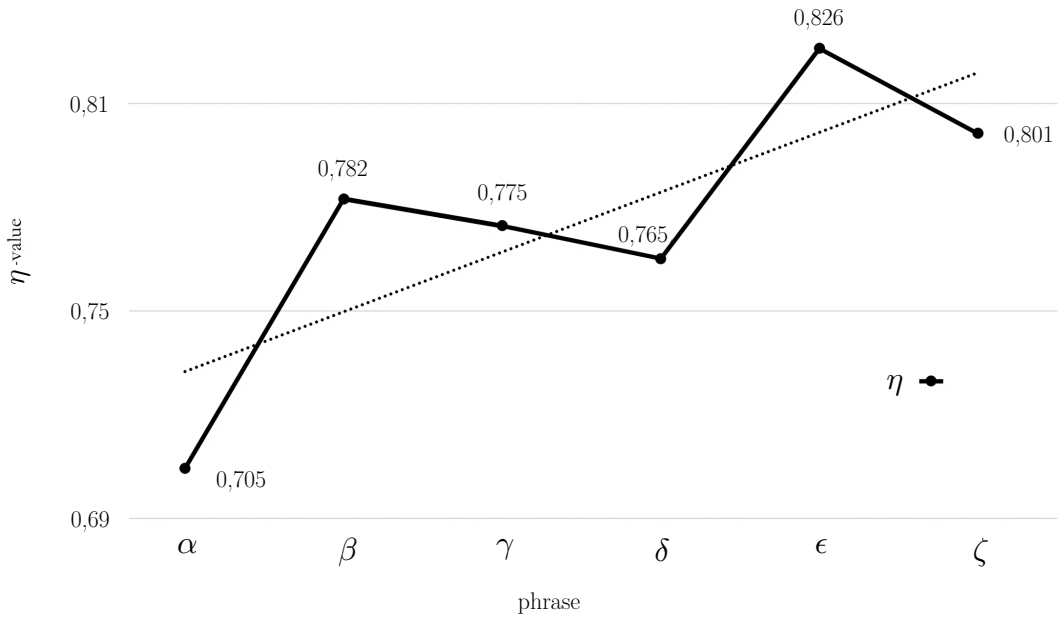
Bipartite and symmetrical structure in Op. 19/1.

3. Contrapuntal linearities and nondirected motion.

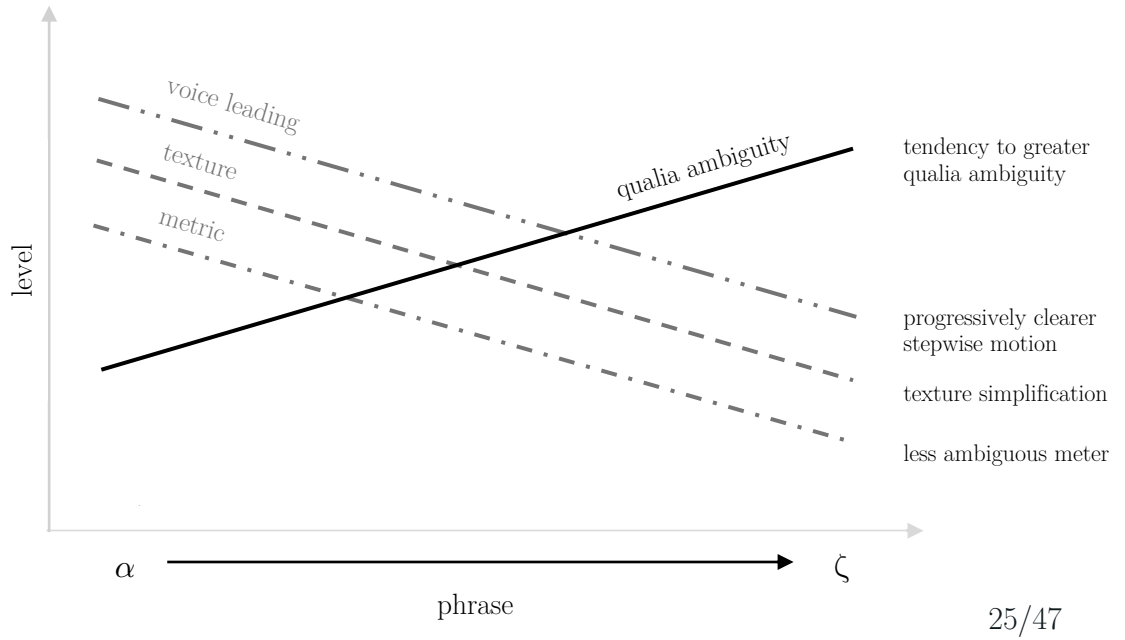
Through the piece, voice-leading, metric, and textural clarity rise, while qualia ambiguity falls — Kramer's nondirected linearity, made measurable.

Ambiguity counterpoints voice-leading, texture, and metre as an **independent musical parameter**.

Evolution of qualia ambiguity in Op. 19/1.



Nondirected and contrapuntal linearities in Op. 19/1.



V. Application 2.

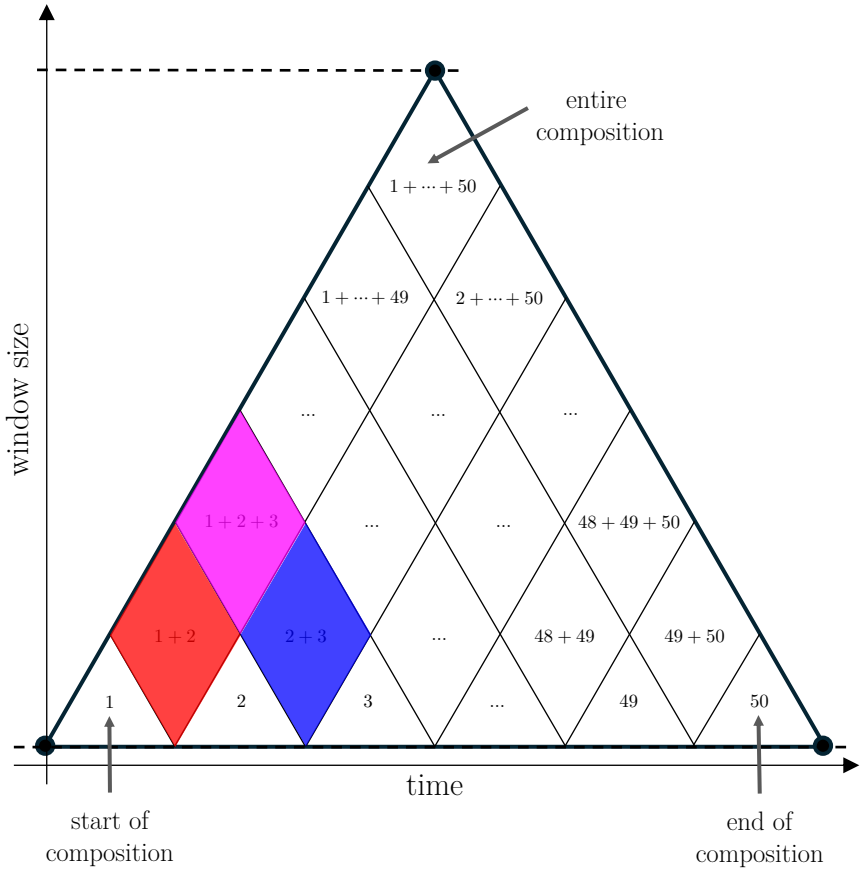
Qualia hierarchies through *wavescapes*.

"if it all builds up less as sound than as stones
sculpted into white and golden volutes whose
shady recesses are a trompe-l'oeil
so that they appear to be more than they are in curved walls;"

Jorge de Sena, in ***Bach: Goldberg Variations***

1. Wavescape structure and colour-mapping the FQS.

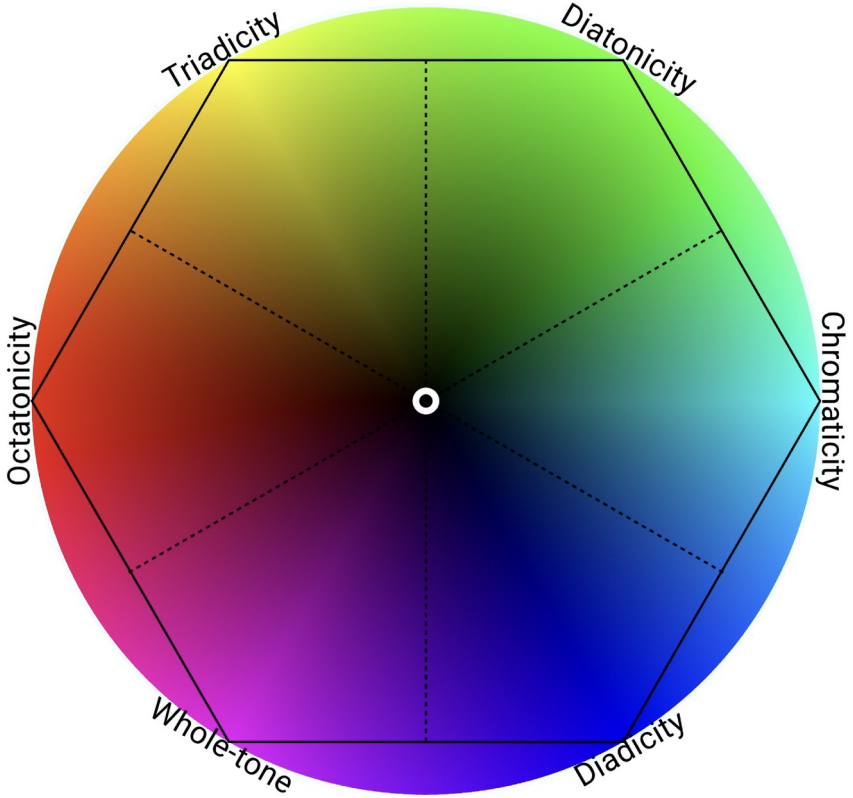
- Wavescape triangle: 50 segments divide a piece at the base and rise to a single apex for the whole piece.
- The wavescape computes the FQS position of each segment and assigns a colour.



Example of a wavescape structure.

1. Wavescape structure and colour-mapping the FQS.

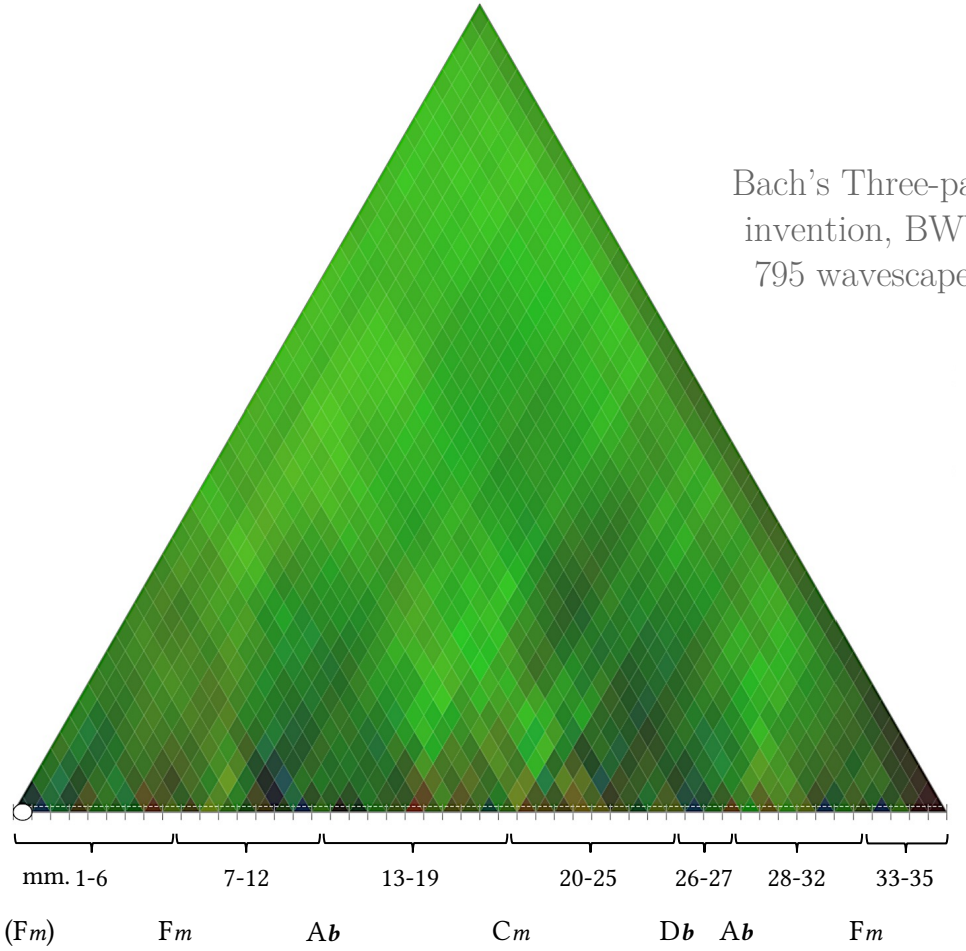
- Wavescape triangle: 50 segments divide a piece at the base and rise to a single apex for the whole piece.
- The wavescape computes the FQS position of each segment and assigns a colour.
- Hue encodes quality around the ring; luminance encodes ambiguity radially.
- Dark means ambiguous; the centre reads black.



Coloured FQS: hue encodes quality, luminance encodes ambiguity.

2. Bach, BWV 795: non structural chromaticity.

- Diatonicity dominates; chromatic, octatonic, triadic, and dyadic hues are local decoration.
- The lament-bass chromaticism is a surface phenomenon inside a fundamentally diatonic (greenish hues) organisation.



Bach's Three-part invention, BWV 795 wavescape.

Listen.

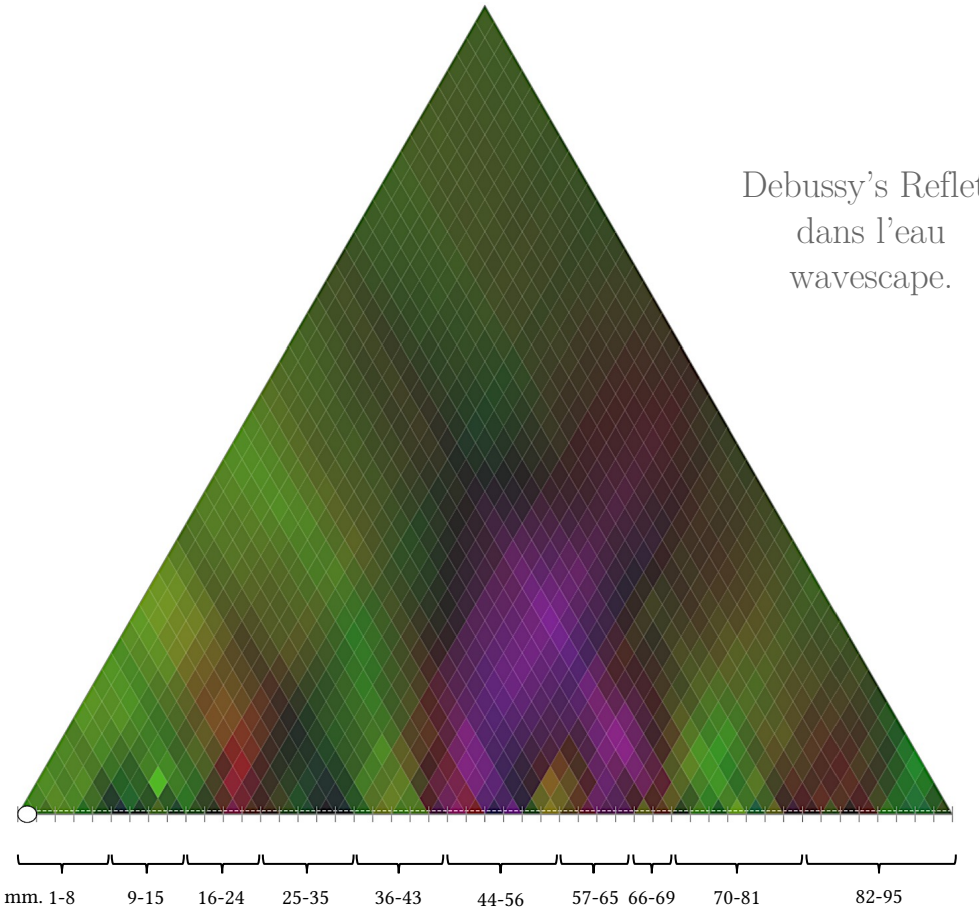
Bach, Three-part invention BWV 795 mm. 1—7.

3. Debussy's Reflets: structural pendullum.

- Pronounced oscillation between diatonic green and whole-tone purple.
- Darker zones reach the middle hierarchies; only a green apex survives.
- Octatonic dominant-ninth sequence in the *Quasi cadenza* (mm. 20–21).

Listen.

Debussy, Reflets dans l'eau, En animant passage, mm. 40—48. Listen 2/3.



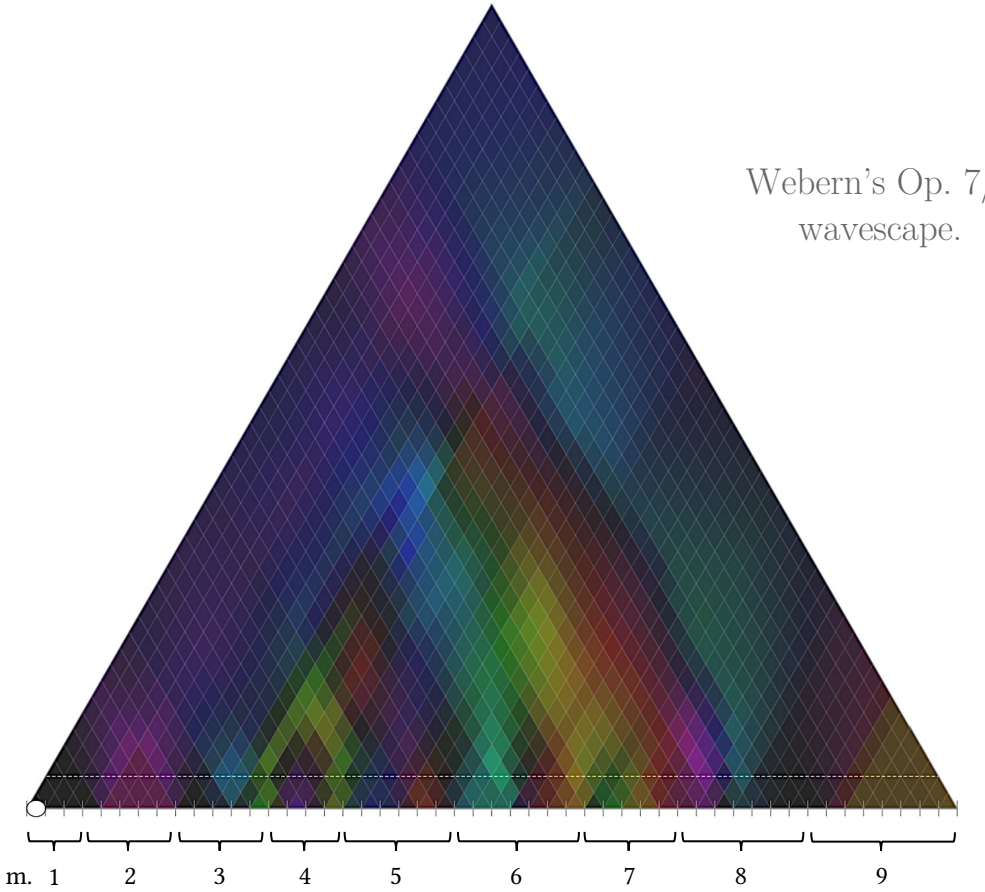
Debussy's Reflets dans l'eau wavescape.

4. Webern, Op. 7/1: qualia pointillism.

- Dominated by blues (dyadic) and purples (whole-tone); kaleidoscopic shifts at every level.
- Qualia plasticity: no quality holds for more than a beat.
- Two nested triangular formations articulate two complete chromatic aggregates.

Listen.

Webern, Op. 7/1.



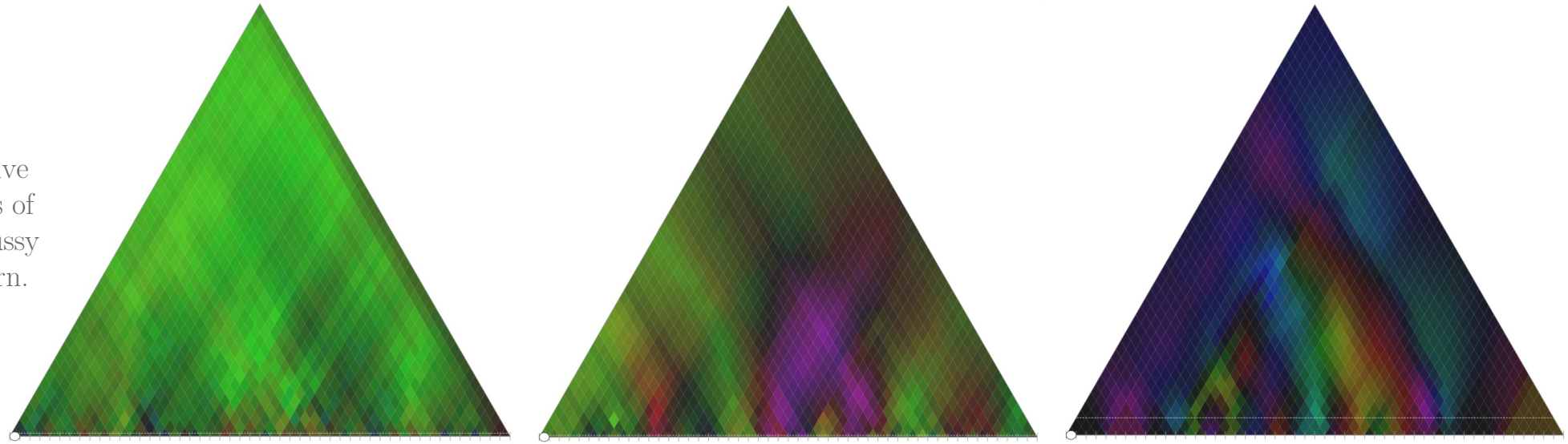
Webern's Op. 7/1 wavescape.

5. Three pieces, three strategies.

- Bach: green throughout; Debussy: pendular green-purple; Webern: kaleidoscopic blues.
- Three pieces don't establish a trend; therefore, it's important to extend the study to an entire corpus.

- Increased qualia ambiguity?

Comparative wavescapes of Bach, Debussy and Webern.



- Bach's Three-Part Invention wavescape.
- Debussy's *Reflets dans l'eau* wavescape.
- Webern's Op. 7, No. 1 wavescape.

1723 — 1905 — 1910-1914

Timeline

VI. Application 3.

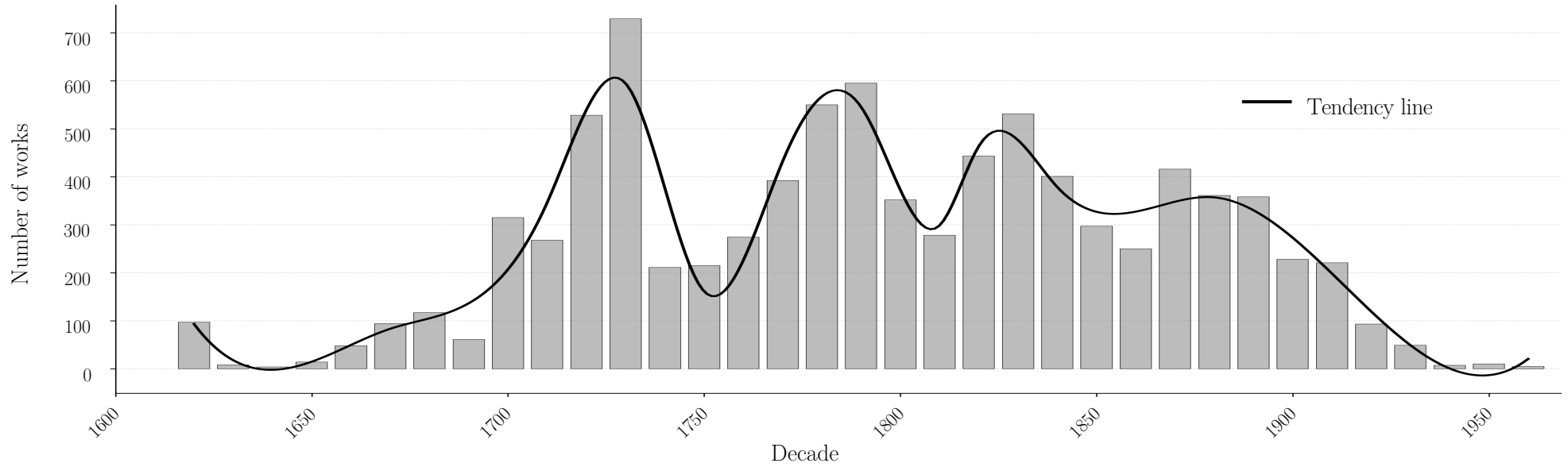
A linguistic turn.

"a power disguising itself in languor, an air of inspiration
concealing the structure, a harmonious melancholy above
an ironic melody (or the other way round), the magic of rhythms
used to hide thought—and hide it so well"

Jorge de Sena, in ***Chopin: An Inventory***

1. The corpus and a closer look at Debussy.

- Extended Yale–Classical Archives corpus.
- 8,822 works; 501 composers; 1548–1968.
- Augmented with 49 modernist works (Schoenberg 24, Webern 15, Messiaen 9, Berg 1) to address the post-1890 under-representation.



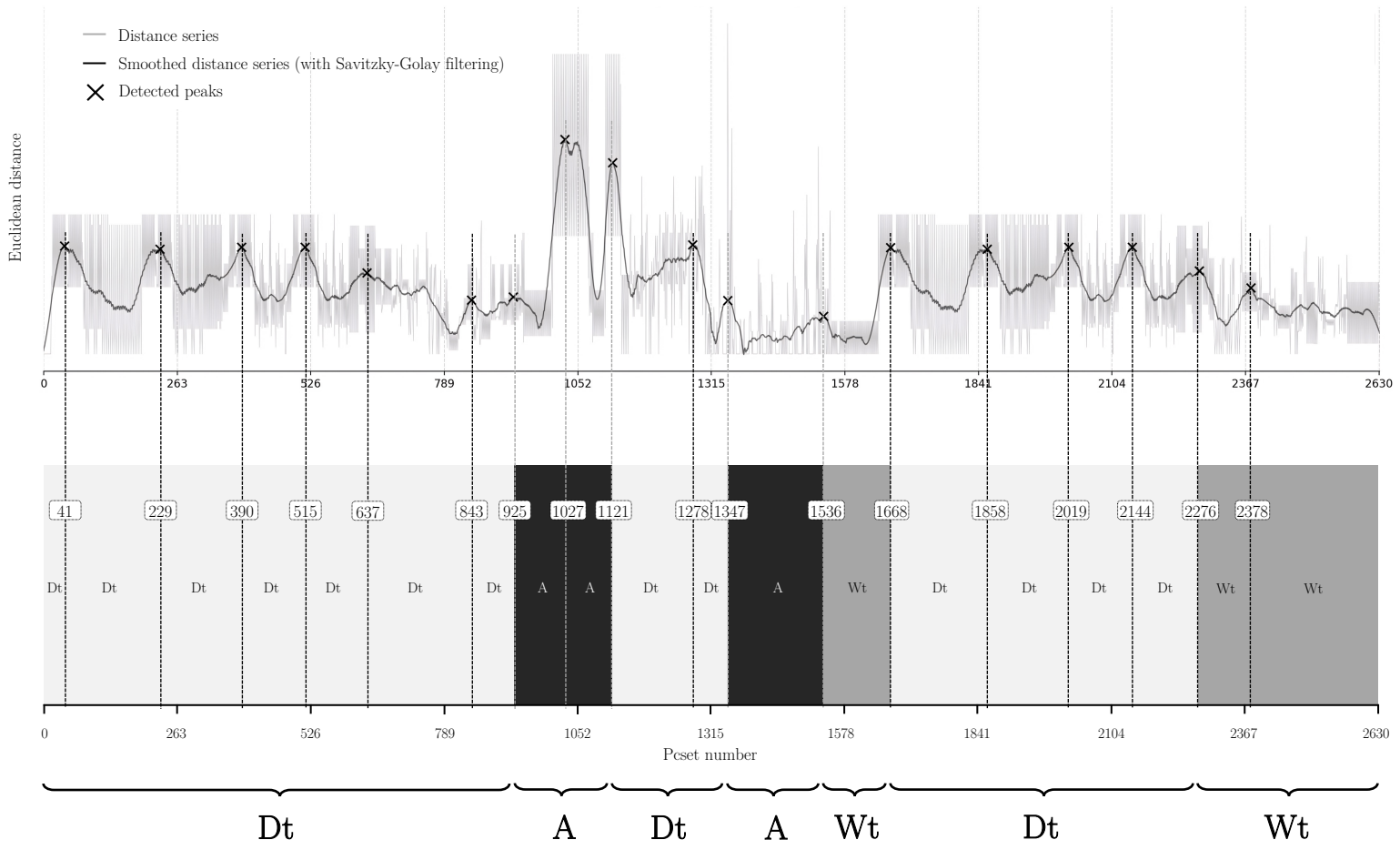
Distribution of works by decade in the extended corpus.

2. Adaptive segmentation.

1. Pitch-class set extraction per slice.
2. DFT → FQS mapping of each slice.
3. For every consecutive triplet (p_{t-1}, p_t, p_{t+1}), compute Euclidean distance $||p_{t+1} - p_{t-1}||$.
4. Savitzky-Golay smoothing of the distance series, followed by prominence-based peak detection.

- Boundaries placed where qualia change, not where measures or phrases change.

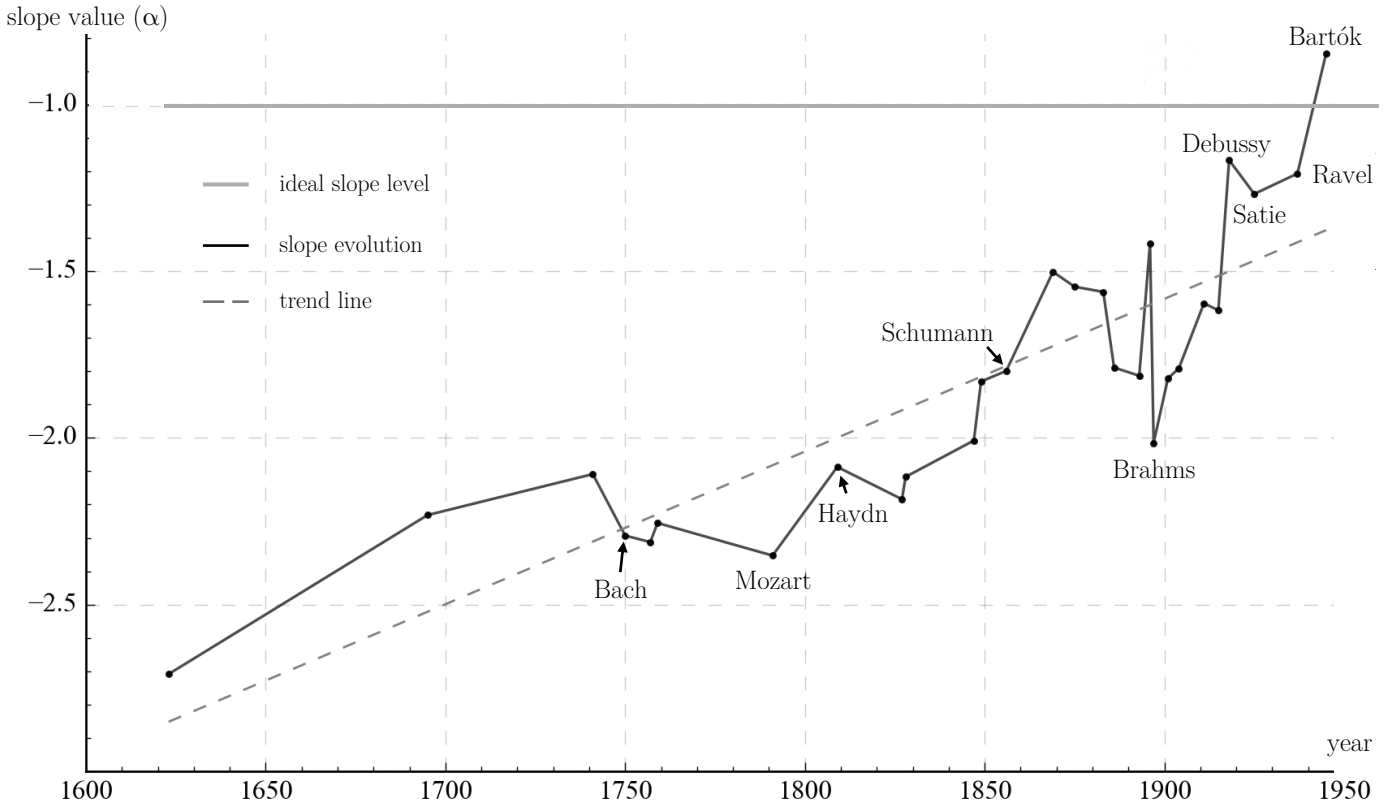
Distance series, segmentation peaks, and qualia progression for Debussy's *Images*, L. 110: No. 3, *Mouvement*.



3.1. Zipf's law and qualia bigrams across composers.

Precedent: Rohrmeier & Cross (2008).

- $f(z) \propto 1/z^{\alpha}$ — the z -th most frequent token appears with a frequency proportional to $1/z^{\alpha}$.
- Natural language: $\alpha \approx -1$. The least-effort distribution.
- Harmonic qualia bigrams become progressively more language-like over four centuries.

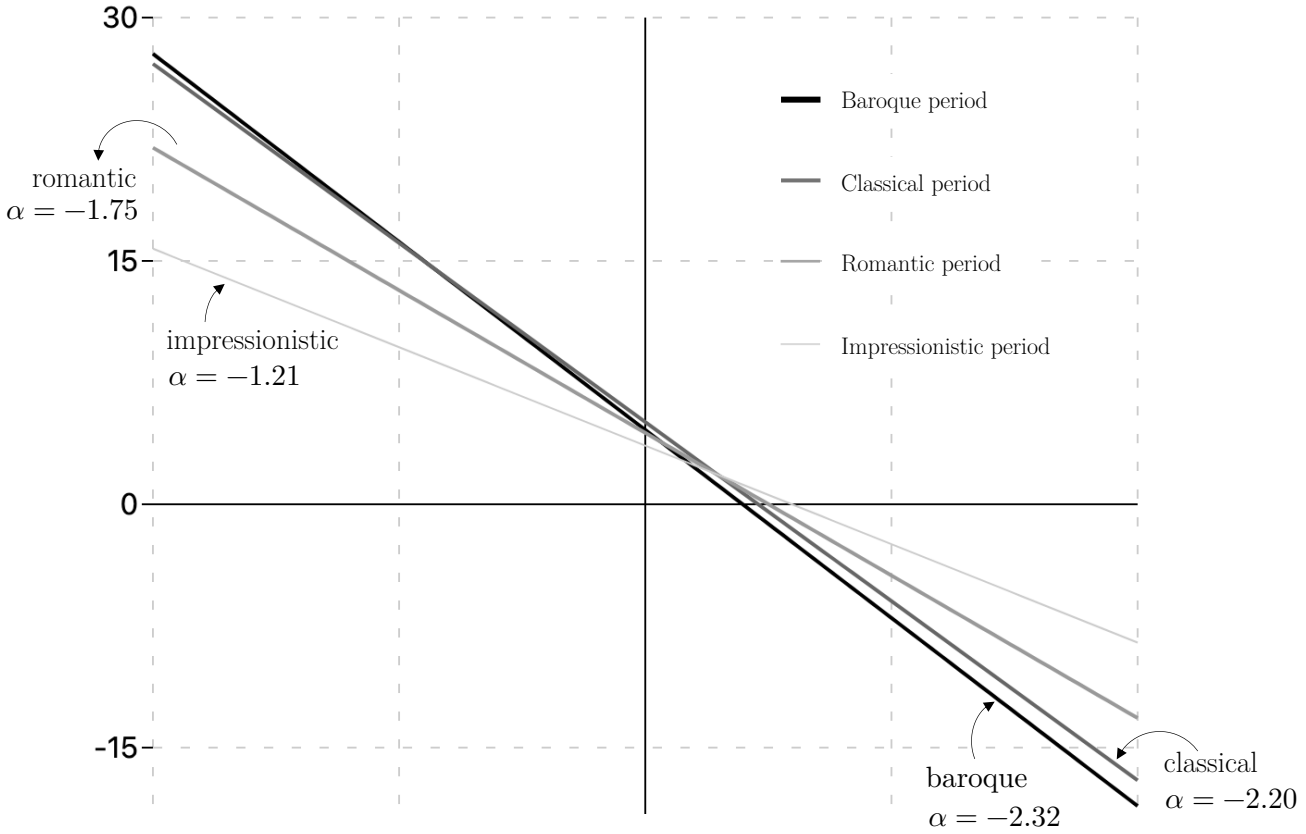


Per-composer log-rank/log-frequency slopes for qualia bigrams.

3.2. Zipf'smean slope by musical period.

- Four periods with one trajectory.
- The closer to $\alpha = -1$, the more language-like the qualia syntax.

	α	Composers
Baroque	-2.32	Byrd, Purcell, Vivaldi, Bach, Scarlatti, Handel
Classical	-2.21	Mozart, Haydn, Beethoven
Romantic	-1.76	Schubert through Mahler (14 composers)
Impressionist	-1.21	Debussy, Satie, Ravel



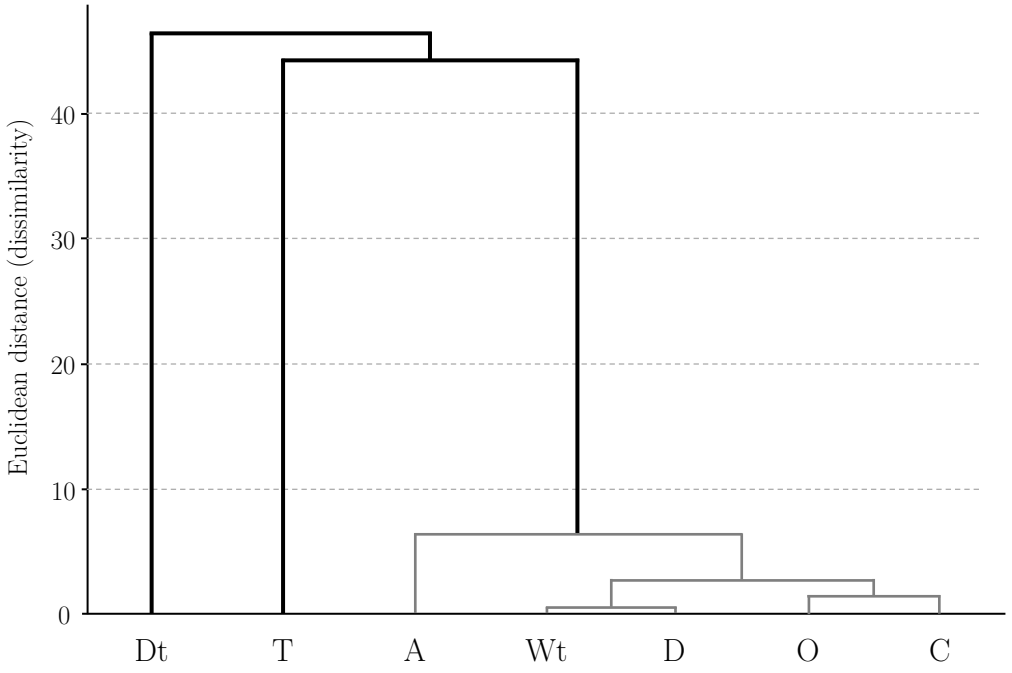
Period-aggregated regressions.

4. From bipartite to tripartite hierarchical clustering.

Hierarchical clustering on transition matrices (Ward's method).

Bach—bipartite

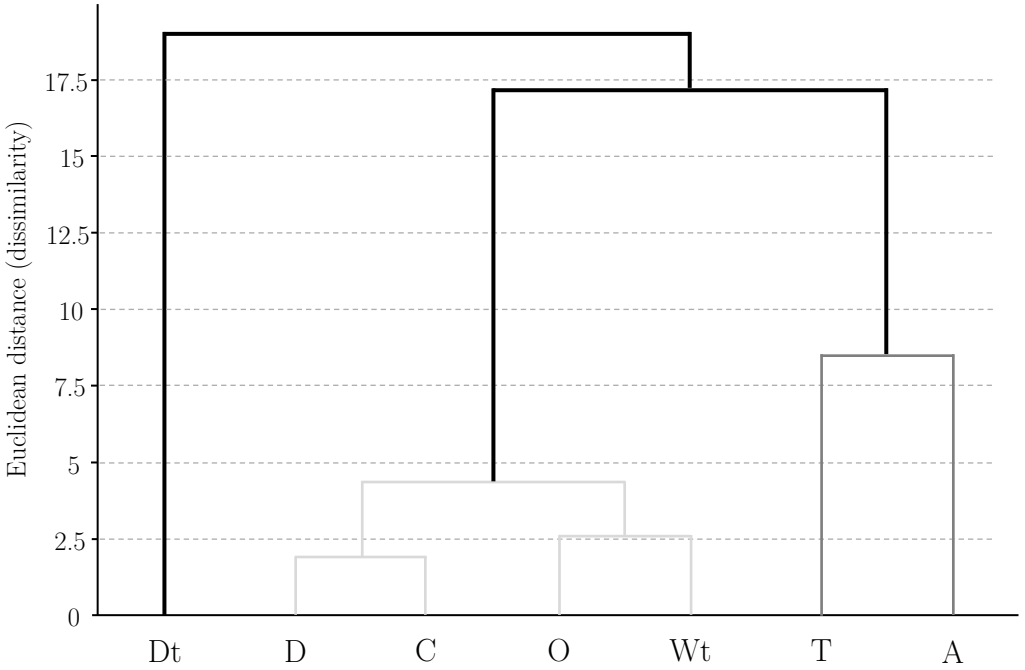
- Cluster 1: Dt + T (tonal core). Cluster 2: all others.



HCA dendrogram on Bach's transition matrices.

Debussy—tripartite

- Dt alone (structural isolation).
- T + A (mediating layer).
- O, Wt, C, D (colouristic periphery).



HCA dendrogram on Debussy's transition matrices.

5. Debussy's three-role model.

Statistical model arising from Debussy's 38 works and 56,193 slices:

Anchor—Diatonicity

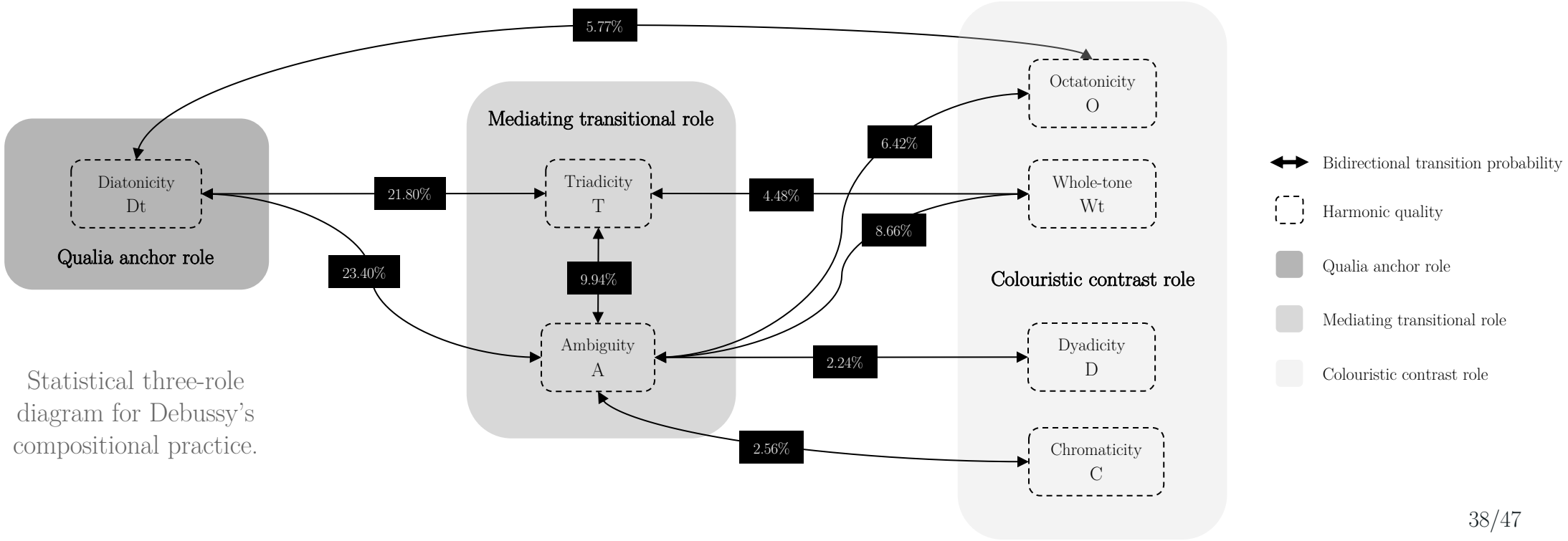
- 44.9% of transitions go to Ambiguity; 61.3% from T return to Dt.

Mediating transitional—T, A

- Ambiguity as hub: 40.7% → Dt, 21% → T, 19.8% → Wt.

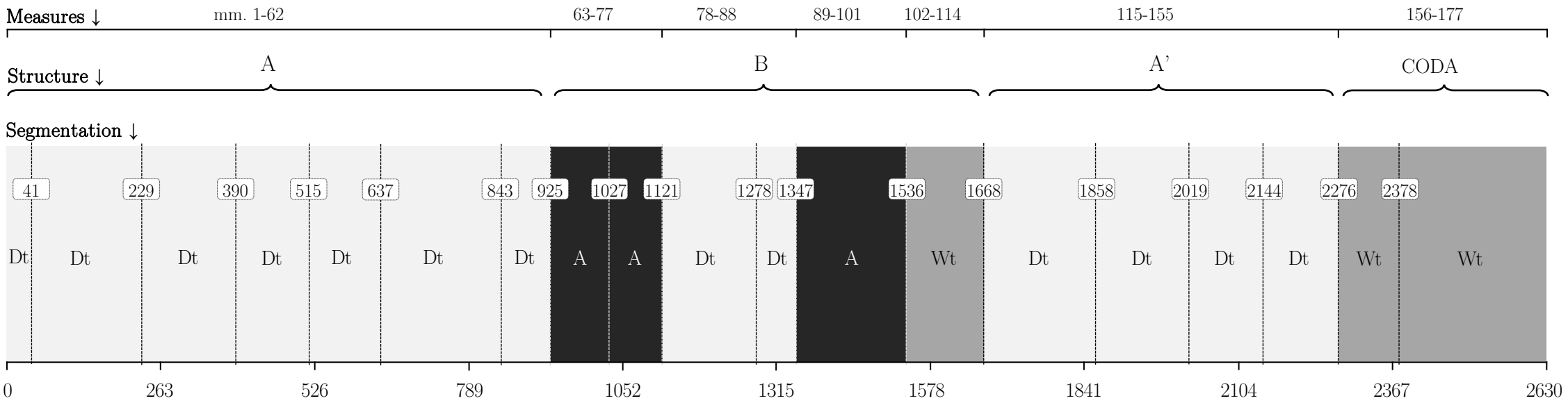
Colouristic contrast—O, Wt, C, D

- Rarely connect directly; 32.67% of trigrams are palindromic (Dt→A→Dt).



Statistical three-role diagram for Debussy's compositional practice.

6.1. *Mouvement*: a qualia progression that mirrors the form.



Roles ↓

QAR
Initial reference point;
Mainly major scales, pentatonic lines, and Lydian mode;
Short chromatic nuances.

MTR
Bridge into whole-tone quality;
Octatonic, chromatic, and whole-tone;
Brief interruption with diatonicity.

CCR
Contrast
Tension.

QAR
Initial reference point re-exposure;
Qualia tension resolution.

CCR
Final qualia contrast;
Unresolved tension.

Progression ↓

Dt → A → Dt → A → Wt → Dt → Wt →

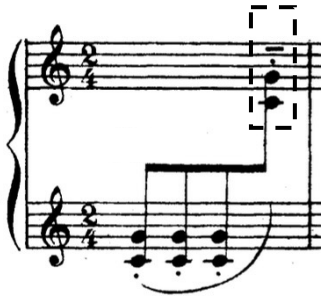
Formal and qualia analyses across Debussy's *Images, Livre* 1: III. *Mouvement*.

6.2. *Mouvement*: why the model captures it?

Diatonic {C,D,E,F,G}/octatonic subsets {C,E♭,F♯,G,A}. The interruption is brief but precisely placed under each tenuto—the articulation reinforces the qualia.

m. 1 ↓

perfect fifth
{0,7}
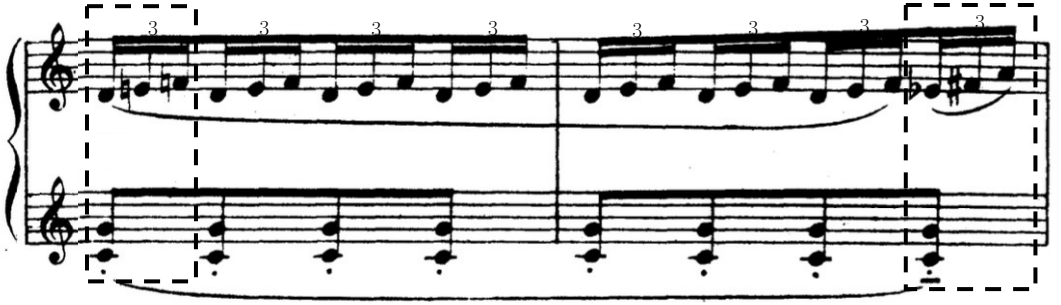


mm. 5-6 ↓

diatonic subset
{0,2,4,5,7}

pitch material

octatonic subset
{0,3,6,7,9}



Articulation patterns: 1. 2. 3. 4__ 1. 2. 3. 4. 5. 6. 7. 8____

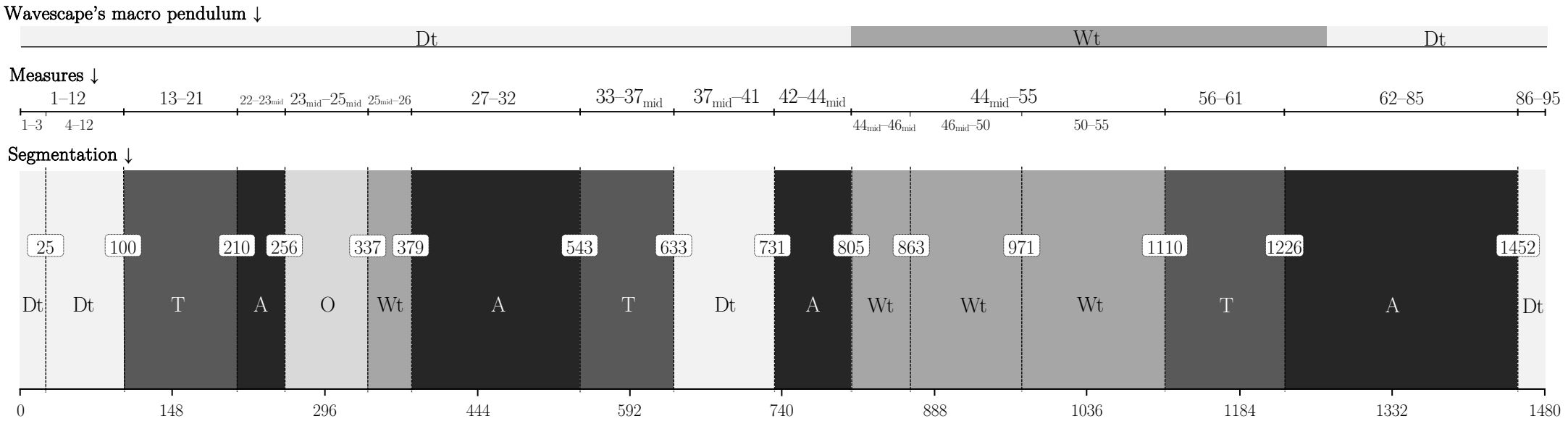
Articulation: 3+1, then 7+1. The pattern of an emphasised last element is mimicked by the qualia progression ending on an unresolved whole-tone CODA.

Listen.

Debussy, *Mouvement* mm. 1—41.

The rhythmic, articulative, and pitch organisation patterns creating the sense of "*movement*" in Debussy's Images, L. 110: No. 3.

7.1. *Reflets dans l'eau* through the model.



Progression ↓

Dt → T → A → O → Wt → A → T → Dt → A → Wt → T → A → Dt

Roles ↓

QAR
- First qualia anchor.
- Gravitational pull.

MTR
- Triadicity arising from joint octatonic, pentatonic, and triadic elements.
- Ambiguity defined by two chromatic melodic lines in contrary motion.

CCR
- Strongest departure from qualia stability.
- Introducing contrast and tension.

MTR
- Articulates diatonic, triadic, and chromatic elements.
- Facilitates the return to qualia anchor role.
- Transitional character emphasised by Tempo slowing down.

QAR
- Second qualia anchor.
- Resolution of qualia tension.
- Starts another cycle.

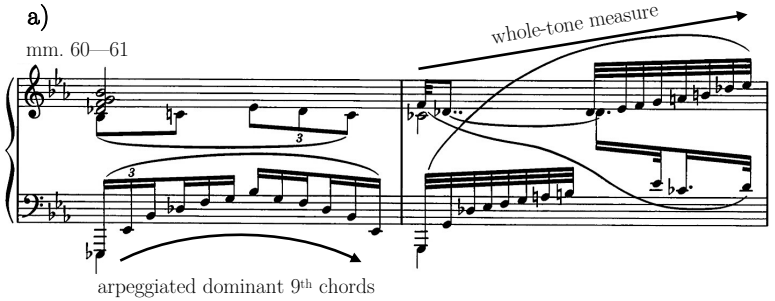
MTR
- New transition passage.
- Different context.
- Distinct role.

CCR
- Mainly whole-tone material;
- Second and final moment of colouristic contrast in the work.

MTR
- Work's climax (m. 57).
- Multiple harmonic shifts.
- Articulates diatonic, triadic, and chromatic elements.
- Facilitates the return to qualia anchor role.

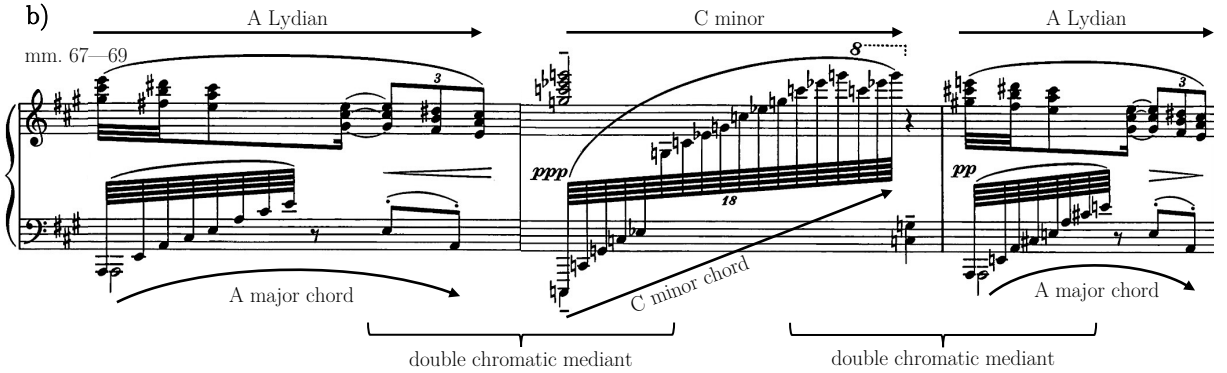
QAR
- Final qualia rest.
- Slicing slow down.
- Final resolution of qualia tension.

7.2. Inside the mediating layer: *Reflets'* structured ambiguity.

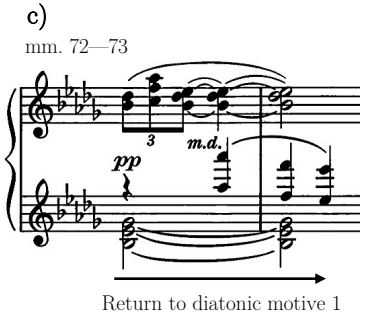


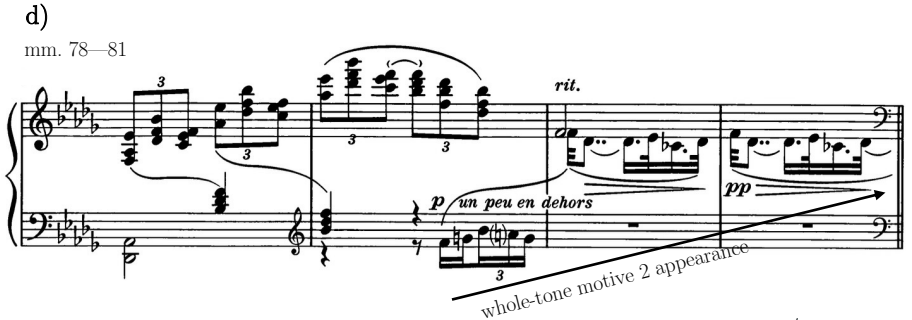
Four examples of qualia moments defining mm. 62—85's segment as ambiguous.

- Inside the ambiguous segment, a microcosm of the three-role model.



Listen. Debussy, *Reflets dans l'eau* mm. 64—71. Listen 3/3.





VII. Postlude.

Synthesis and conclusion.

"Music is no longer the same, and we're no longer the same.
Not because time flows, or because the dome rises
forever between us and our selves. No. It's because
the virtual thought became
an evidence: became concrete."

Jorge de Sena, in ***Bach: Goldberg Variations***

1. Three gaps, three answers.

↪

Interaction & Motion

- A convex 2-D geometry in which progressions are continuous trajectories.
- Op. 19 No. 1 becomes a circular path.

?

Ambiguity

- $\eta = 1 - \overline{\ell}$ metric turns qualia ambiguity to a measurable state.
- Qualia ambiguity functions as a nondirected and contrapuntal linear parameter in Schoenberg's Op. 19/1.



Multidimensionality

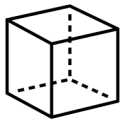
- All six coefficients visualised at once through the RadViz spring-force model.
- Three-role functional model emerges because all six dimensions are present simultaneously.

2. The contributions of the thesis, by type.



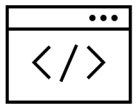
Harmonic qualia reconceived in four dimensions: spatial-geometric, temporal-accumulative, hierarchical, and phenomenological. A measurable, multi-scale construct.

→ ***Conceptual:*** *reconceptualising qualia.*



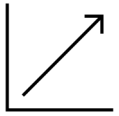
A six-dimensional DFT projection rendered as a single hexagonal layout. Convex, transposition-invariant, and equipped with the qualia ambiguity metric η.

→ ***Methodological:*** *the Fourier Qualia Space.*



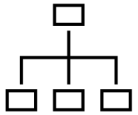
Adaptive distance-sensitive segmentation; centroid-based reference sonority extraction; FQS-coloured qualia wavescapes. Open-source code.

→ ***Technical:*** *segmentation and visualisation tools.*



Op. 19/1 reread: qualia ambiguity as a linear parameter like voice-leading, metric, and texture. Qualia ambiguity reframed as parametric counterpoint.

→ ***Analytical:*** *ambiguity as nondirected linearity.*



Anchor, Mediating, and Colouristic roles recovered statistically. Functional differentiation operating on qualities rather than pitches.

→ ***Theoretical:*** *the three-role model.*

3. Where the framework falters, and what awaits.

Limitations

- The FQS introduces roughly 27% distance distortion.
- The method is ceteris paribus and purely harmonic.
- Statistical patterns in Debussy's three-role model, are emergent tendencies rather than planned practice.
- Potential links between perception along with the inherent properties of DFT-derived harmonic qualities, remain untested.

Future work

- **Theoretical:** extending the model to composers such as Bartók, Ravel, Stravinsky towards a fuller account of qualia syntax.
- **Technical and analytical:** refining the segmentation for new repertoires and extending the results to other works.
- **Pedagogical:** the space's visual, intuitive form makes it a promising way to teach post-tonal harmony.

Thank you!

Selected references

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